

0 2

X-rays and ultrasound are used in medical imaging.

0 2

1

State **one** advantage of using X-rays instead of ultrasound in medical imaging.

[1 mark]

0 2

2

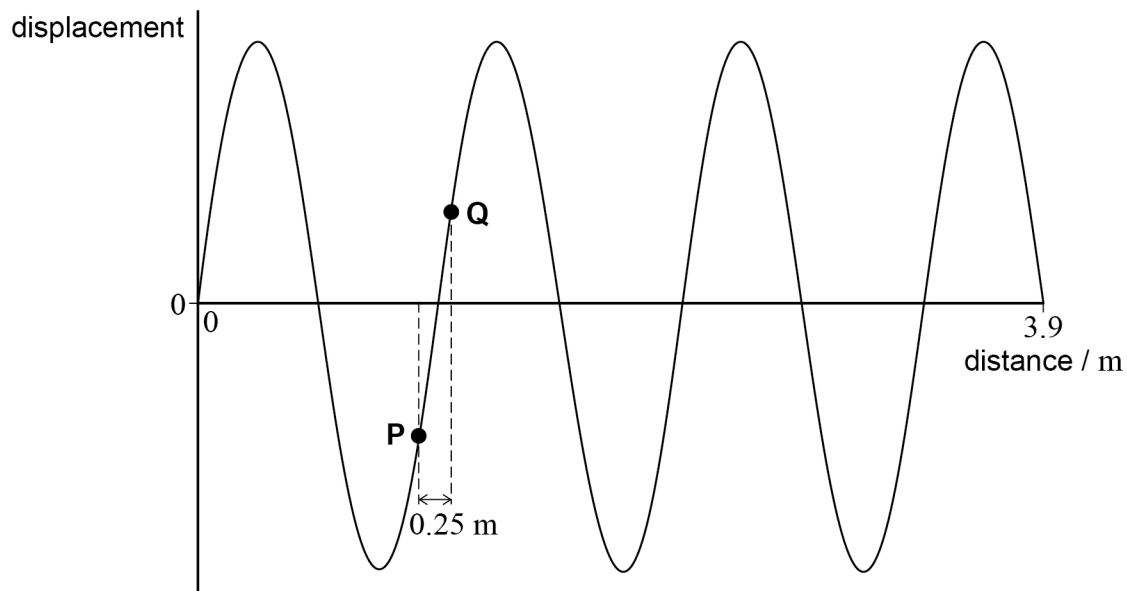
Describe why ultrasound, rather than X-rays, is used to produce an image of a fetus.

[2 marks]



Do not write

The graph shows the variation with distance of the displacement of a progressive wave.



What is the phase difference between points **P** and **Q**?

[1 mark]

A 0.22 rad

☐

B 0.40 rad

☐

C 1.2 rad

☐

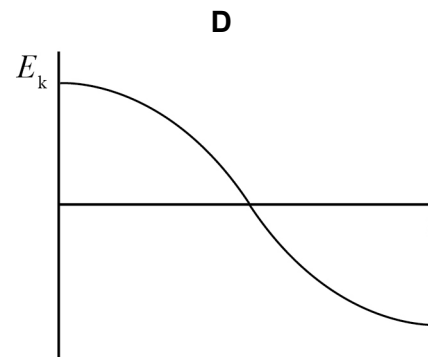
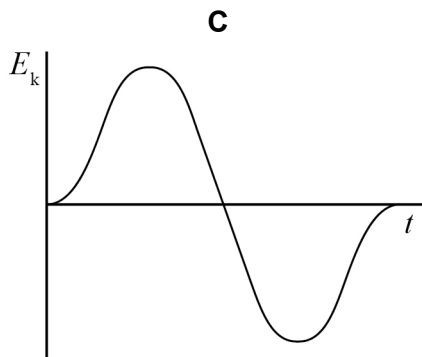
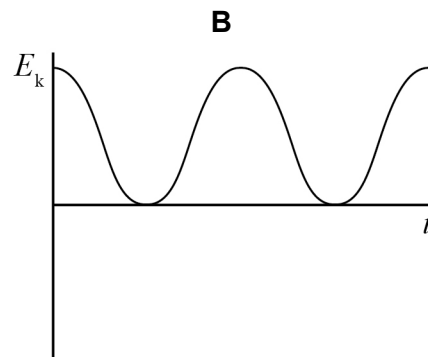
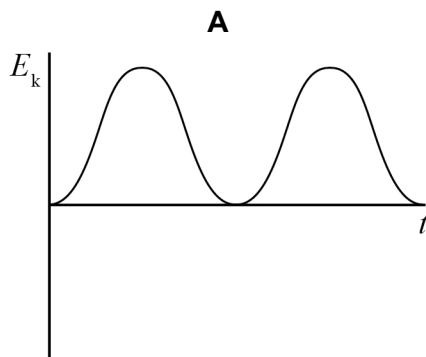
D 1.4 rad

☐

A simple pendulum is released from its extreme position and completes one oscillation.

Which graph shows the variation of kinetic energy E_k with time t for the oscillation?

[1 mark]



- A** ☐
- B** ☐
- C** ☐
- D** ☐

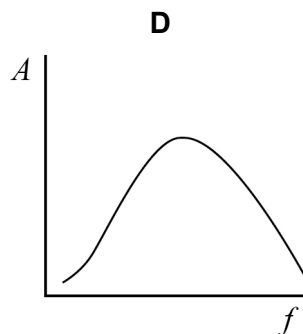
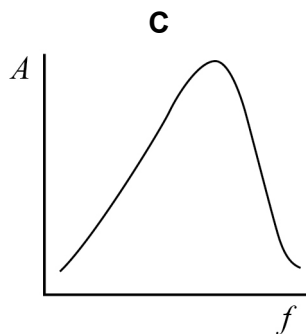
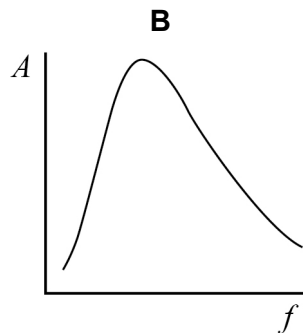
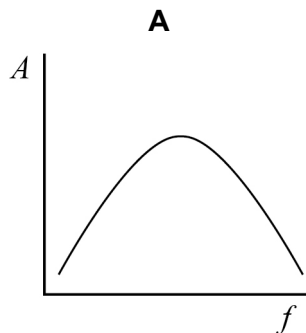


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2 3

Which graph shows the variation of amplitude A with driving frequency f for a damped system undergoing forced oscillations?

[1 mark]

A ☐B ☐C ☐D ☐

0	1
---	---

Explain why longitudinal waves cannot be polarised.

[2 marks]

2

0

1

An electron is travelling at $1.29 \times 10^6 \text{ m s}^{-1}$.

Calculate its de Broglie wavelength.

[2 marks]

wavelength = _____ m

2

0	3
---	---

A student is provided with the following apparatus:

- a lamp
- a polarising filter
- a light meter that measures the intensity of incident light.

Describe how the student could use this apparatus to determine whether the light from the lamp is polarised.

[3 marks]

3

Turn over for the next question

Turn over ►





0 4

A washing machine is loaded unevenly. When the machine is turned on, the drum rotates with an increasing frequency. At one particular frequency the system vibrates with a large amplitude. As the frequency increases further, the amplitude of the vibrations decreases.

0 4 . 1

Explain why the large amplitude vibrations occur.

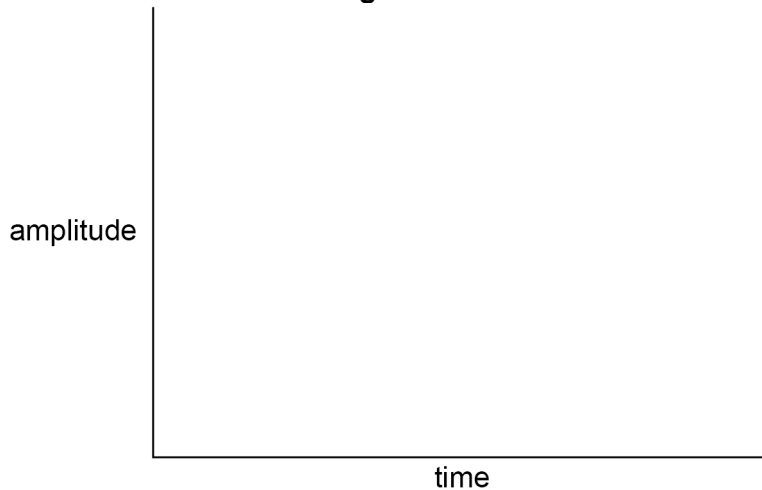
[3 marks]

0 4 . 2

Sketch on **Figure 1** the variation with time of the amplitude of vibrations as the frequency of the drum's rotation increases.

[1 mark]

Figure 1



0 4 . 3

Explain how increasing the damping of the system affects the graph you sketched in question **04.2**.

[2 marks]



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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Figure 7 shows a circuit used to determine the emf and internal resistance of a battery.

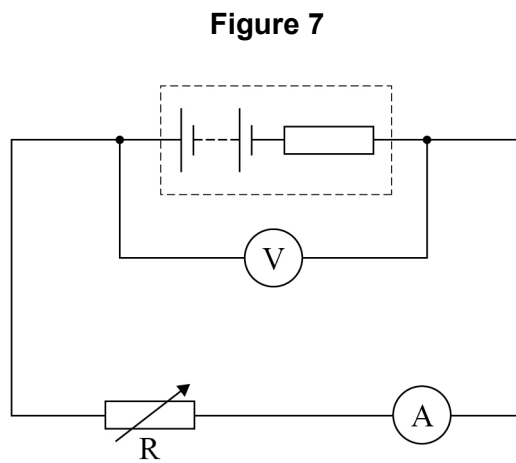
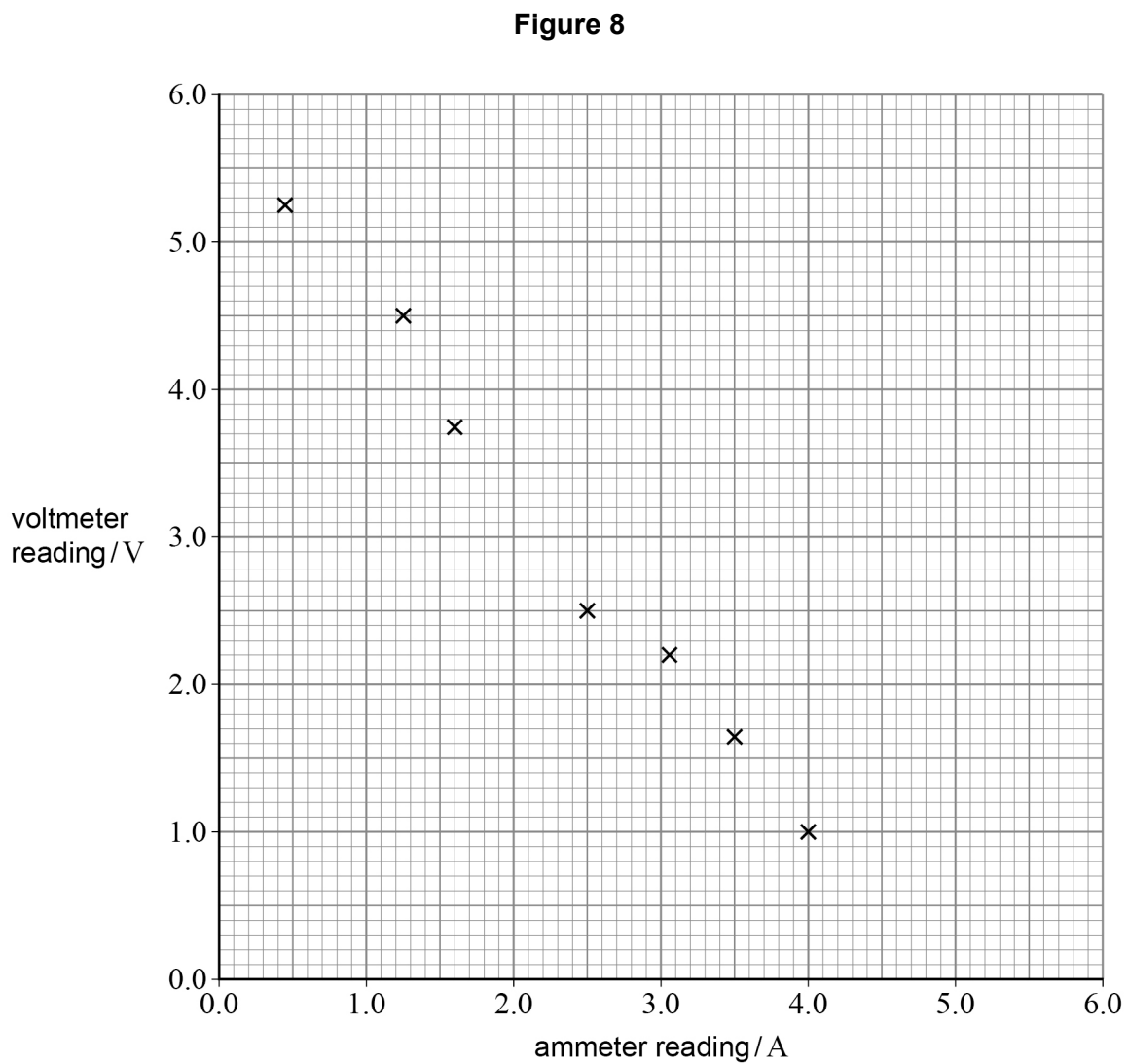


Figure 8 shows data from this experiment.



0 7 . 1 Explain why the voltmeter reading decreases as the current increases.

[2 marks]

0 7 . 2 Determine the emf of the battery.

[2 marks]

emf = _____ V

0 7 . 3 Determine the internal resistance of the battery.

[3 marks]

internal resistance = _____ Ω

0 7 . 4 A second battery has half the emf and half the internal resistance of the first battery.

Draw a line on **Figure 8** to show the variation of voltmeter reading with ammeter reading for the second battery.

[2 marks]



0 8 . 1 Progressive waves transfer energy along the wave; stationary waves do not.

Describe **two other** differences between stationary waves and progressive waves.

[2 marks]

Difference 1 _____

Difference 2 _____

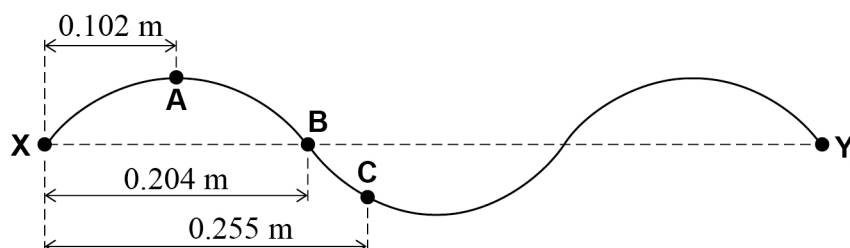
A string of mass 0.98 g vibrates between two fixed ends **X** and **Y**.

The distance between **X** and **Y** is 0.612 m.

The tension in the string is 69 N.

Figure 9 shows one position of the string when vibrating at the third harmonic.

Figure 9



0 8 . 2 Determine the frequency of the third harmonic.

[4 marks]

frequency = _____ Hz



0 8 . 3 Calculate the speed of the waves on the string.

[3 marks]

speed = _____ m s^{-1}

0 8 . 4 Point **B** is shown on **Figure 9**.

Describe the behaviour of the string at point **B**.

[1 mark]

0 8 . 5 Points **A** and **C** are shown on **Figure 9**.

State the phase difference between the motion of the string at points **A** and **C**.

[1 mark]

phase difference = _____ rad

END OF SECTION A



Section B

Answer **all** questions in this section.

0 9

Light undergoes total internal reflection in an optical fibre with no cladding.
The critical angle of light in this fibre without cladding is 43.6° .

0 9

1

Show that the speed of light in the fibre is approximately $2.1 \times 10^8 \text{ m s}^{-1}$.

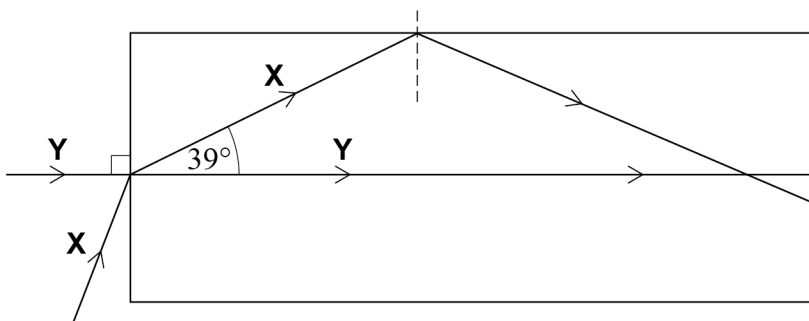
[2 marks]



0 9 . 2

Figure 10 shows the paths followed by two pulses of light, **X** and **Y**, entering the straight optical fibre at the same time. **X** travels through the fibre at 39° to the axis of the fibre. **Y** travels along the axis of the fibre.

Figure 10



The fibre is 320 m long.

Calculate the time delay between pulse **Y** leaving the fibre and pulse **X** leaving the fibre.

[4 marks]

time delay = _____ s

Question 9 continues on the next page

Turn over ►



0

9

.

3

Modern optical fibres are made with cladding around the core.

Explain how using cladding reduces pulse broadening.

[2 marks]

8



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1	0
---	---

A student determines the period T of oscillation for a mass–spring system by measuring ten oscillations of the system. The measurements were repeated.

Table 1 shows these measurements.

Table 1

$10T / \text{s}$	12.63	12.60	12.73	12.80	12.71
------------------	-------	-------	-------	-------	-------

1	0
---	---

1

Calculate the mean value for T .

[1 mark]

mean value for $T =$ _____

1	0
---	---

2

Calculate the percentage uncertainty in your mean value for T .

[1 mark]

percentage uncertainty in $T =$ _____



10.3 The mass m on the spring was (0.400 ± 0.008) kg.

Calculate the spring constant k .

[1 mark]

$$k = \text{_____} \text{ N m}^{-1}$$

10.4 Calculate the percentage uncertainty in your answer for k .

[2 marks]

$$\text{percentage uncertainty in } k = \text{_____}$$

Question 10 continues on the next page

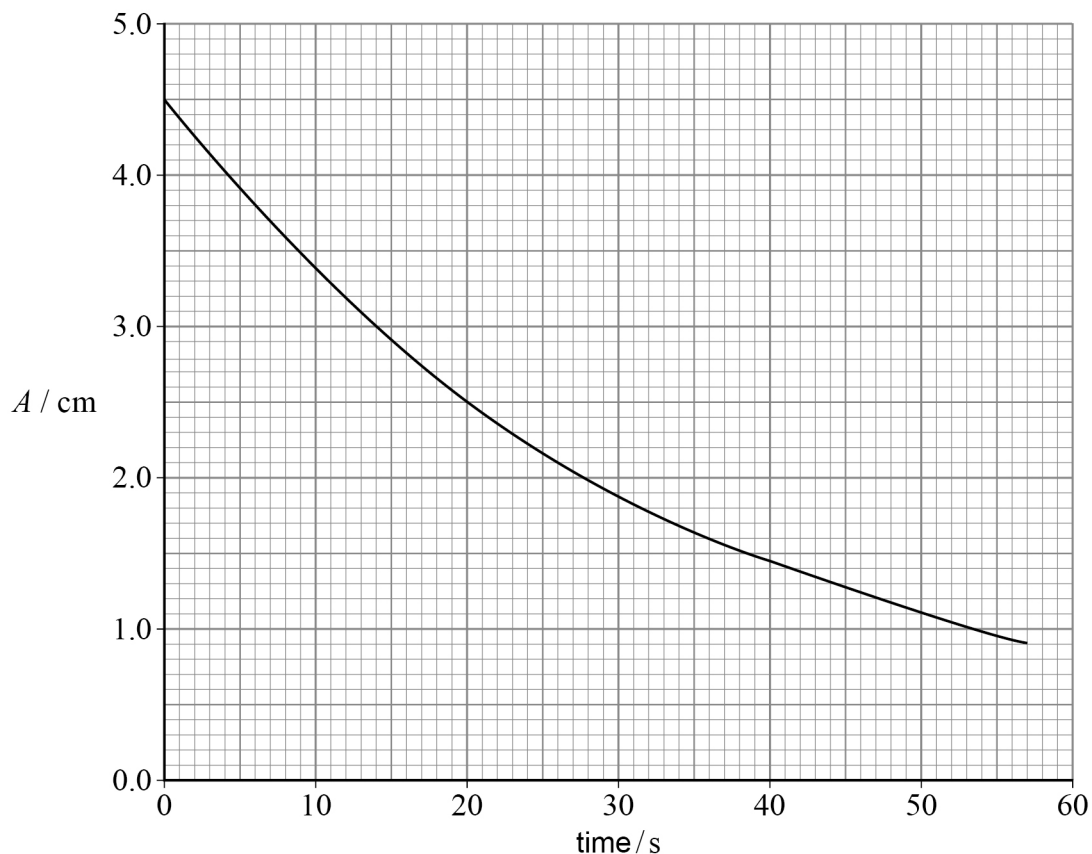


1 0 . 5

To investigate damping in the system, the amplitude A of the oscillations was measured until a total of fifty oscillations were completed.

Figure 11 shows the variation of amplitude with time.

Figure 11



The damping in the mass–spring system causes the amplitude to decrease. The time taken for the amplitude of this system to decrease by half is constant, and is called the half-life.

Determine a reliable value for the half-life of this system.

[3 marks]

half-life = _____ s

END OF SECTION B

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS

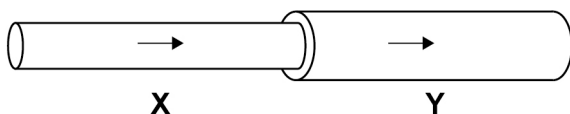


If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

- 1 1** Two wires, **X** and **Y**, are connected in series.
X and **Y** are made of the same material and have the same length. The diameter of **Y** is larger than the diameter of **X**.



Which statement is correct?

[1 mark]

- A** The resistance of **X** is smaller than the resistance of **Y**. ☐
- B** The current in **X** is smaller than the current in **Y**. ☐
- C** The pd across **X** is equal to the pd across **Y**. ☐
- D** The power dissipated by **X** is larger than the power dissipated by **Y**. ☐

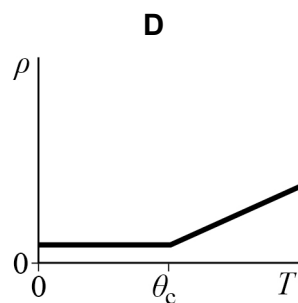
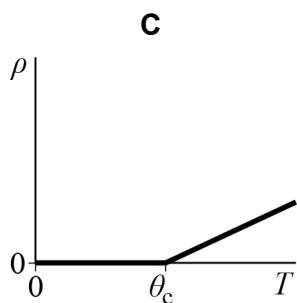
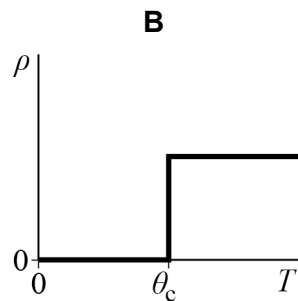
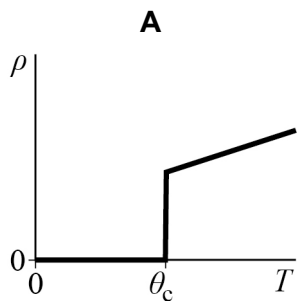


1 2

A superconductor has critical temperature θ_c .

Which graph shows the variation of resistivity ρ with temperature T for this superconductor?

[1 mark]



A ☐

B ☐

C ☐

D ☐

Turn over for the next question

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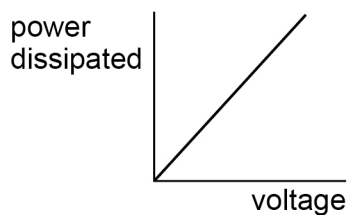
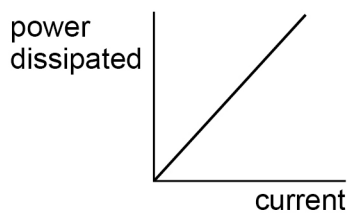


1 3

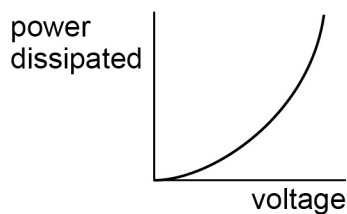
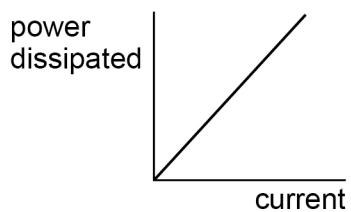
Which pair of graphs shows the variation of power dissipated with current, and the variation of power dissipated with voltage, for a resistor of constant resistance?

[1 mark]

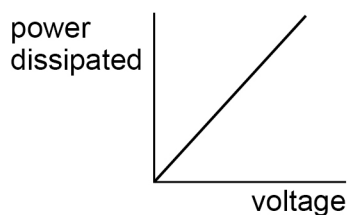
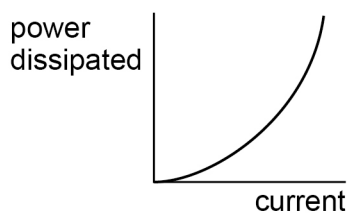
A


☐

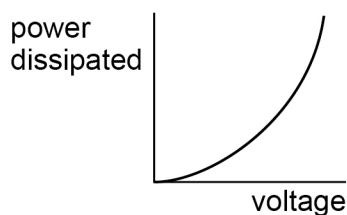
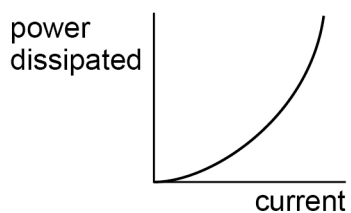
B


☐

C

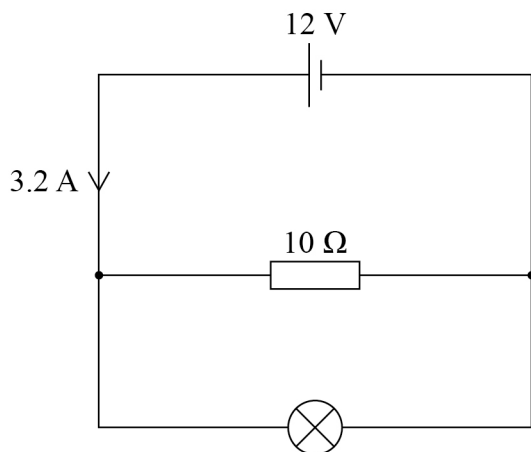

☐

D


☐


1 4

A cell of negligible internal resistance is connected to a resistor and a lamp in parallel as shown.



What is the power dissipated by the lamp?

[1 mark]

A 14 W

☐

B 16 W

☐

C 24 W

☐

D 38 W

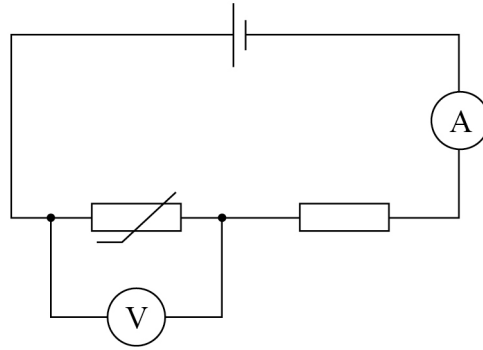
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Turn over for the next question

Turn over ►



1 5 A negative temperature coefficient thermistor is connected to a resistor and a cell as shown.



The temperature of the thermistor increases.

What are the changes in the ammeter reading and the voltmeter reading?

[1 mark]

	Ammeter reading	Voltmeter reading	
A	decreases	decreases	☐
B	decreases	increases	☐
C	increases	decreases	☐
D	increases	increases	☐



1 6 A pendulum of length l oscillates at frequency f .

Which length of pendulum oscillates at frequency $2f$?

[1 mark]

A $\frac{l}{4}$ ☐

B $\frac{l}{2}$ ☐

C $\frac{l}{\sqrt{2}}$ ☐

D $l\sqrt{2}$ ☐

1 7 A stationary wave forms on a string when progressive waves are reflected at the fixed ends. For a real string, energy is transferred from the progressive waves when they are reflected at the fixed ends.

The effect of this energy transfer on the stationary wave is that

[1 mark]

A the antinodes and nodes no longer form. ☐

B the nodes are not positions of zero amplitude. ☐

C the distance between antinodes increases. ☐

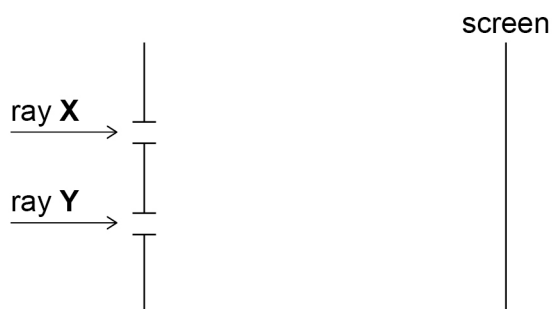
D the speed of the waves on the string is decreased. ☐

Turn over for the next question

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- 1 8** Ray **X** is incident on one slit and ray **Y** is incident on another slit of a double slit arrangement as shown below.



A fringe pattern is **not** observed on the screen.

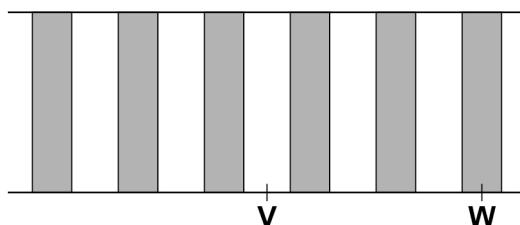
What is a possible explanation?

[1 mark]

- A** **X** and **Y** are not incident at 90° to the slits.
- B** **X** and **Y** do not have the same wavelength.
- C** **X** and **Y** are too bright.
- D** **X** and **Y** are out of phase at the position of the slits.

☐
☐
☐
☐

- 1 9** Light of wavelength 500 nm is incident on a double slit and the resulting fringe pattern is shown below. **V** is the central bright fringe and the dark fringes are shown shaded.



What is the path difference that results in fringe **W**?

[1 mark]

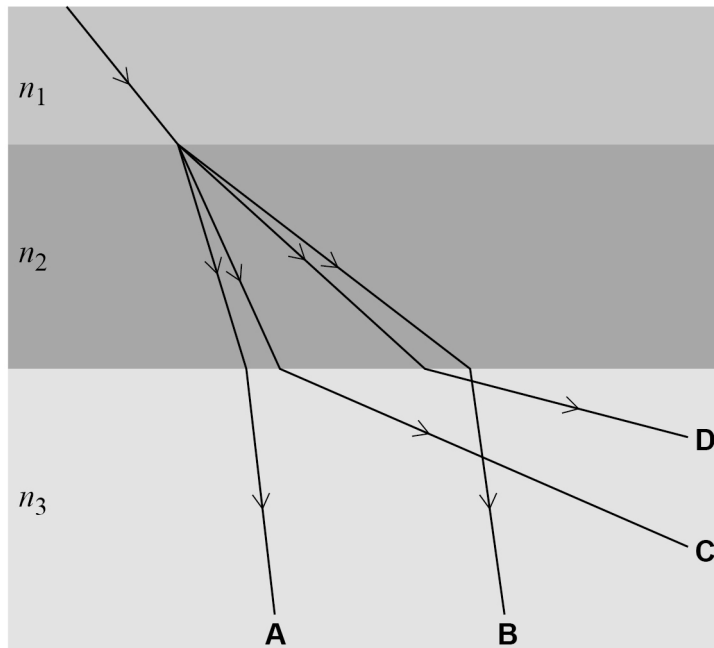
- A** 1250 nm
- B** 1000 nm
- C** 750 nm
- D** 500 nm

☐
☐
☐
☐


2 0

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A ray of light travels through three materials, with refractive indices n_1 , n_2 and n_3 , where $n_3 < n_1 < n_2$.



Which is a possible path for the ray?

[1 mark]

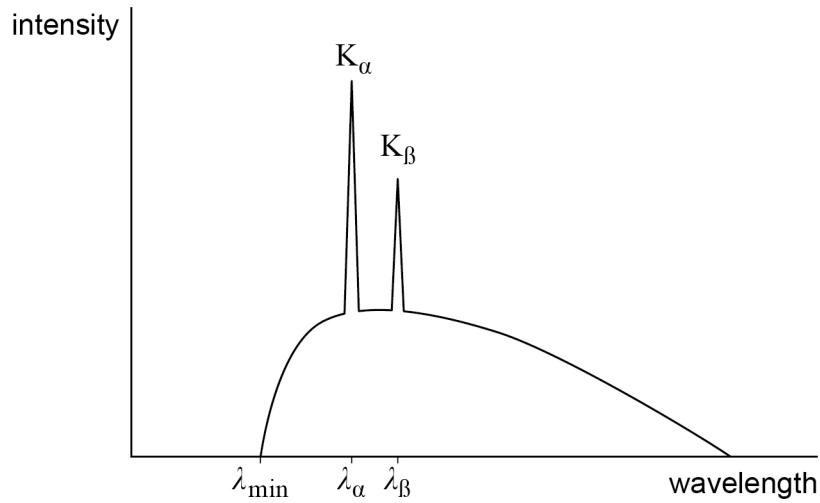
- A ☐
- B ☐
- C ☐
- D ☐

Turn over for the next question

Turn over ►



2 1 The graph shows the X-ray spectrum emitted by an X-ray tube with a tungsten target, operated at 50 kV.



The same X-ray tube, with the same tungsten target, is now operated at 100 kV.

What happens to the minimum wavelength $\lambda_{\min}$ of the spectrum and the wavelengths λ_{α} and λ_{β} of the characteristic K-lines K_{α} and K_{β} ?

[1 mark]

	$\lambda_{\min}$	λ_{α}	λ_{β}	
A	halves	halves	halves	☐
B	doubles	doubles	doubles	☐
C	halves	unchanged	unchanged	☐
D	doubles	unchanged	unchanged	☐



2 2

In the photoelectric effect, what is the relationship between the work function ϕ of a material and its stopping potential V_s ?

[1 mark]

A $\phi = hf - V_s e$

☐

B $\phi = V_s h$

☐

C $\phi = \frac{hc}{V_s}$

☐

D $\phi = hf + V_s e$

☐

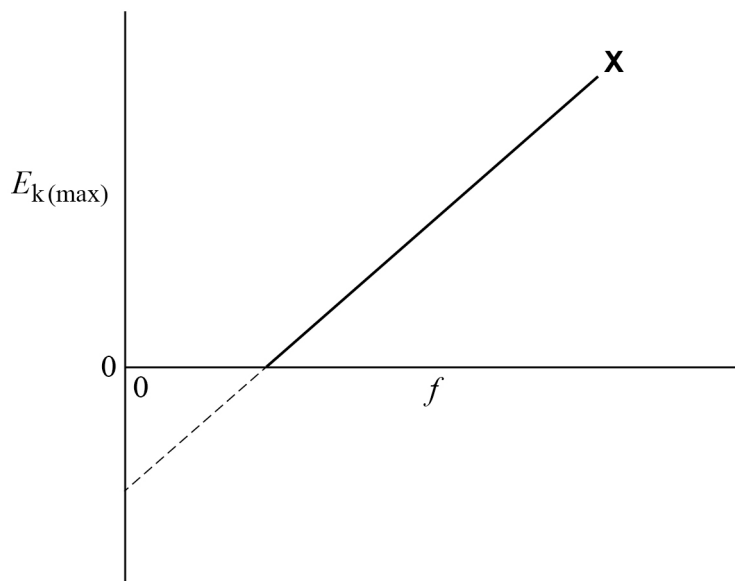
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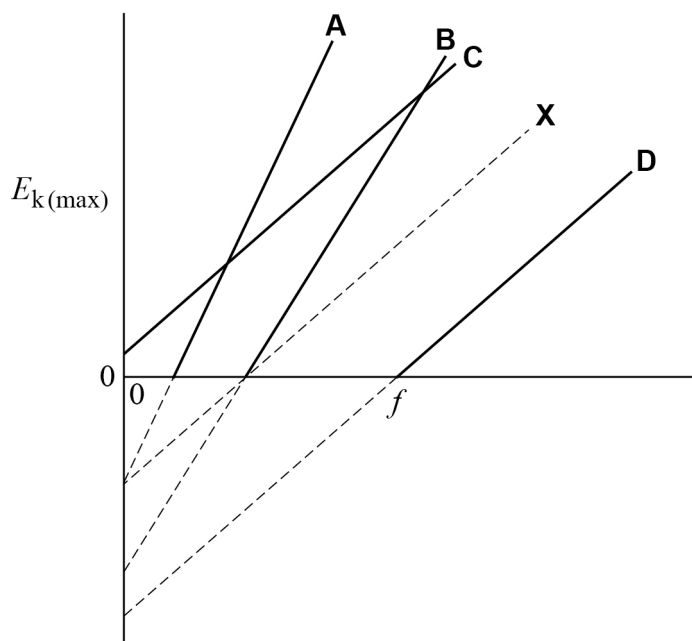
2 3

The graph shows the relationship between the maximum kinetic energy $E_{k(\max)}$ of photoelectrons emitted from material **X**, and the frequency f of incident light.



Which line shows the variation of $E_{k(\max)}$ with f for a different material?

[1 mark]



A

☐

B

☐

C

☐

D

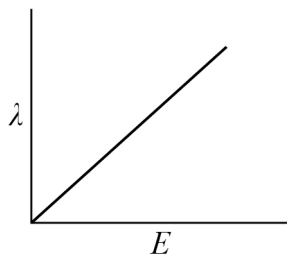
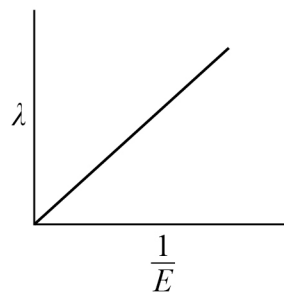
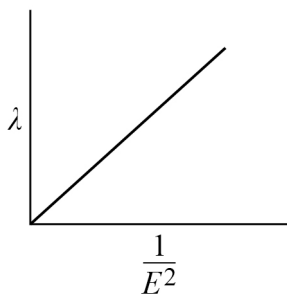
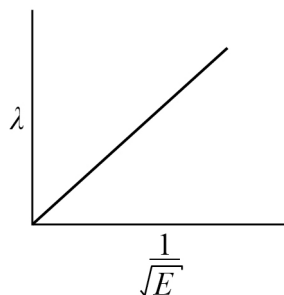
☐


2 4

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Which graph shows the relationship between the de Broglie wavelength λ of an electron and its kinetic energy E ?

[1 mark]

A**B****C****D****A**☐**B**☐**C**☐**D**☐

[illegible]

[illegible]

[illegible]

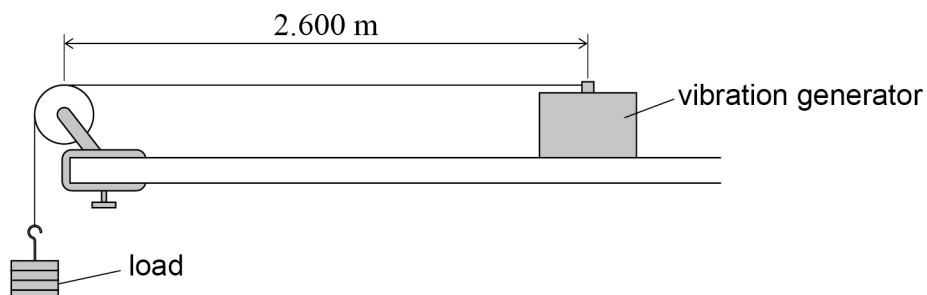
[illegible]

[illegible]

0 5 **Figure 4** shows the apparatus used to investigate stationary waves on a string.

The frequency of the vibration generator is adjusted until the first-harmonic stationary wave is observed on the string. This is repeated for different loads.

Figure 4



0 5 . 1 Show that the frequency f of the first harmonic is related to the tension T in the string by the following equation:

$$f^2 = \left(\frac{1}{4lm} \right) T$$

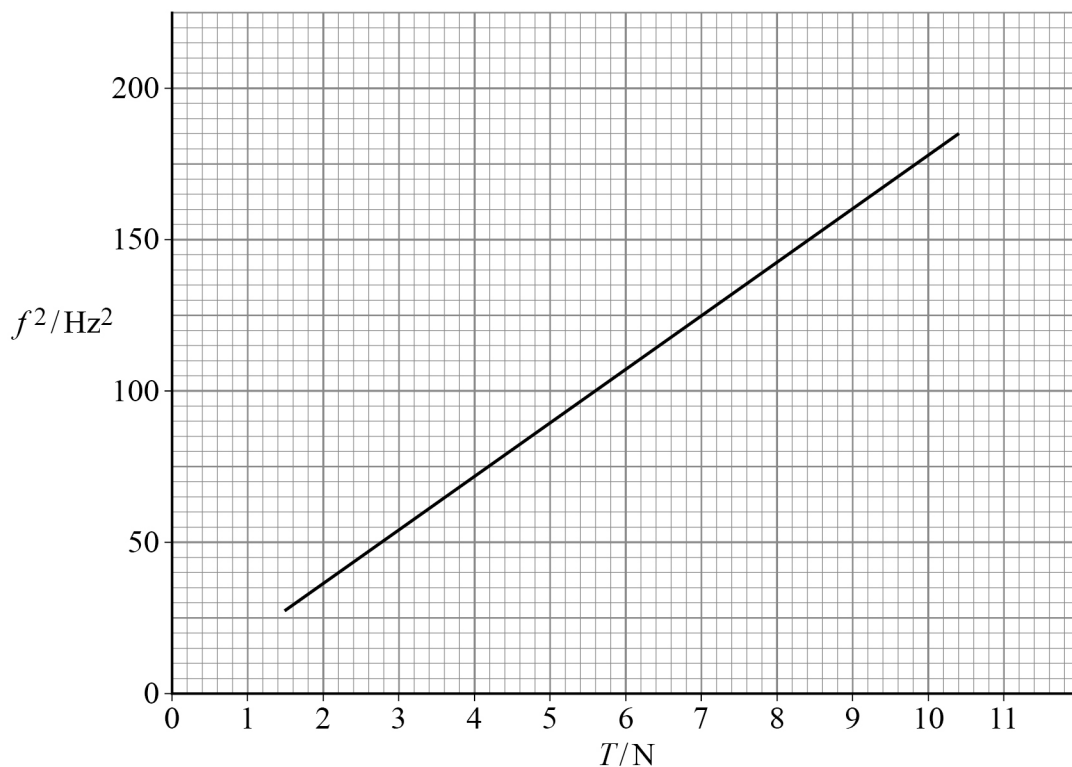
where l = length of the vibrating part of the string
 m = mass of the vibrating part of the string.

[2 marks]



0 5 . 2 Figure 5 is a graph showing the variation of f^2 with T for this investigation.

Figure 5



Length l is 2.600 m.

Determine, using **Figure 5**, an accurate value for m .

[3 marks]

$m =$ _____ kg

Question 5 continues on the next page

Turn over ►



0 5 . 3

To measure the length of 2.600 m, a metre ruler or a 5 m tape measure may be used. Both the metre ruler and the tape measure have a 1 mm resolution.

Explain the advantage of using the 5 m tape measure.

[2 marks]

7



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A string is stretched between two fixed points. It is plucked in the middle so that a stationary wave forms on the string.

Explain how the stationary wave is formed.

[3 marks]



Don't write

State and explain **one** piece of evidence that shows that energy levels in atoms are discrete.

[3 marks]

Turn over for the next question



0 **5** . **1**

Show that an X-ray of frequency 7.60×10^{17} Hz has a wavelength of approximately 4.0×10^{-10} m.

[1 mark]

0 **5** . **2**

Show that an electron travelling at a speed of 4.10×10^6 m s⁻¹ has a de Broglie wavelength of approximately 1.8×10^{-10} m.

[1 mark]

0 **5** . **3**

A student is investigating the atomic spacing in graphite. The atomic spacing is approximately 0.15 nm.

The two possible methods are:

- using the diffraction of the X-rays in Question **05.1**
- using the diffraction of the electrons in Question **05.2**.

State and explain which is the better method.

[2 marks]



0 6 . 1

An electric current is passed through a fluorescent tube.

Electrons and ions collide with mercury gas in the fluorescent tube, raising the mercury atoms to higher energy levels.

Describe how a fluorescent tube produces visible light.

[3 marks]

0 6 . 2

The resistance of the tube decreases when the potential difference across it is increased.

Suggest how this happens.

[2 marks]



0 7

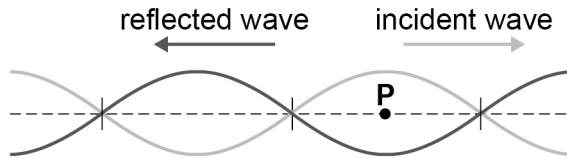
A wave travels along a stretched string and is reflected from a fixed end. The incident wave and the reflected wave interact to produce a stationary wave on the string.

Figure 2 shows part of the string, with the incident wave and the reflected wave at time $t = 0$

P is a point on the string.

The incident wave is shown as  and the reflected wave is shown as 

Figure 2



The periodic time for one oscillation of the waves is T .

The amplitude of both the incident wave and the reflected wave is A .

0 7 . 1

State the displacement of the stationary wave at **P** at time $t = 0$

[1 mark]

displacement = _____

Figure 3 shows the **incident** wave at time $t = \frac{T}{4}$.

Figure 3

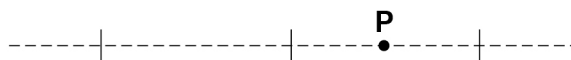


0 7 . 2

Draw on **Figure 4** the **reflected** wave at time $t = \frac{T}{4}$.

[1 mark]

Figure 4



0 7 . 3

Describe and explain the appearance of the stationary wave at time $t = \frac{T}{4}$.

[4 marks]

6

Turn over for the next question

Turn over ►



Figure 5 shows the variation of resistance R_T with temperature for a thermistor.

Figure 6 shows the variation of resistance R_L with light level for an LDR. **Figure 6** is plotted on logarithmic-linear axes.

Figure 5

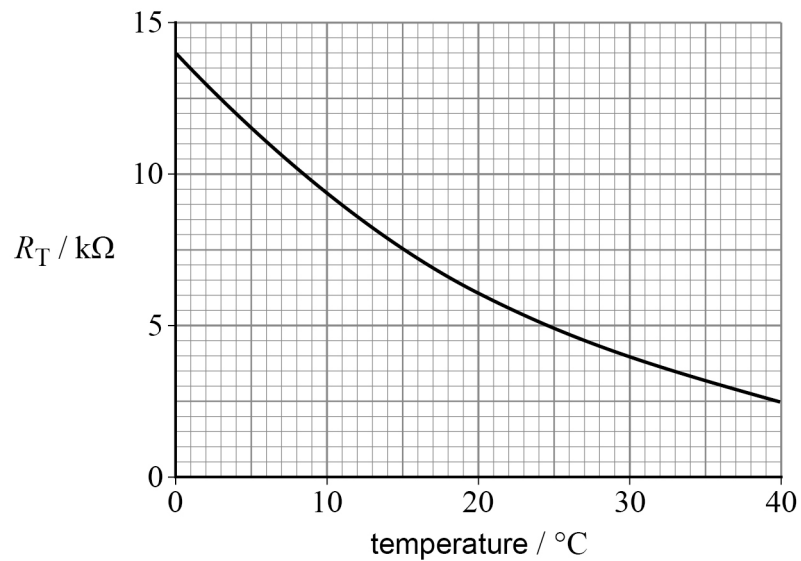


Figure 6

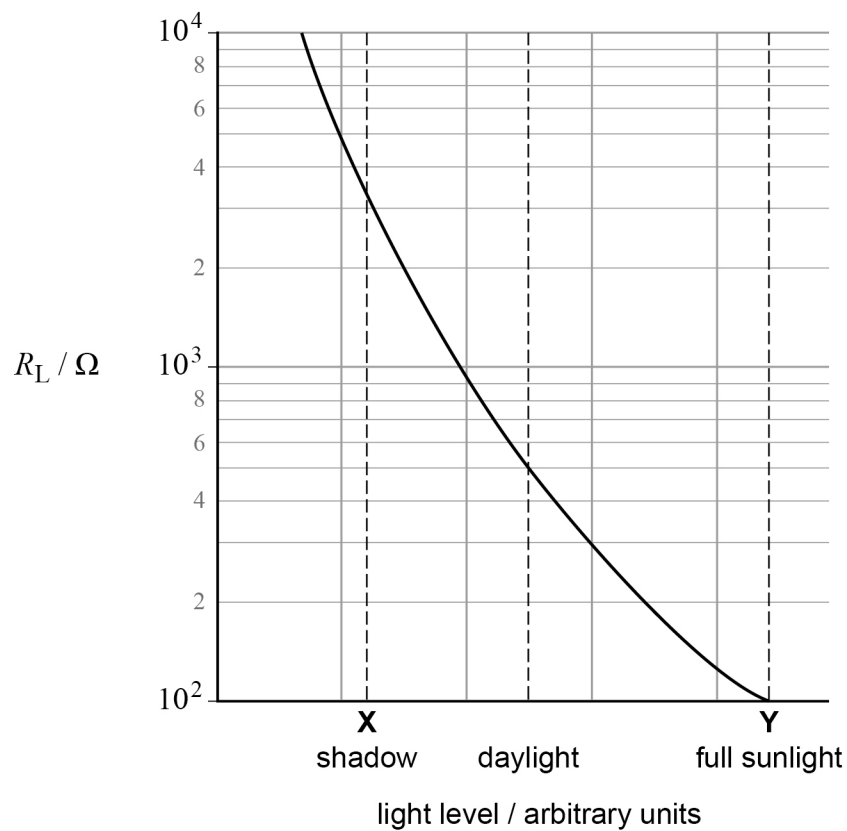
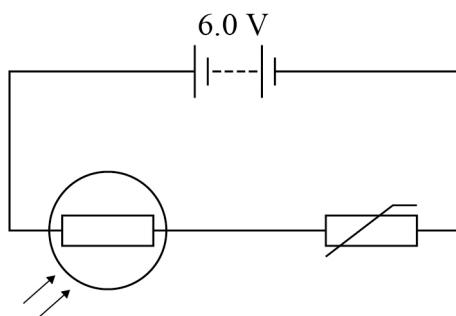


Figure 7 shows a circuit containing the LDR, the thermistor and a 6.0 V battery that has negligible internal resistance.

Figure 7



08.1

The circuit is placed in shadow where the light level is **X** as shown in **Figure 6**. The temperature of the thermistor is 15 °C.

Determine the current in the circuit.

[4 marks]

current = _____ A

08.2

The circuit is now placed in full sunlight where the light level is **Y** as shown in **Figure 6**.

The temperature changes so that the resistance of the thermistor is 5.5 kΩ.

Determine the potential difference across the thermistor.

[3 marks]

potential difference = _____ V

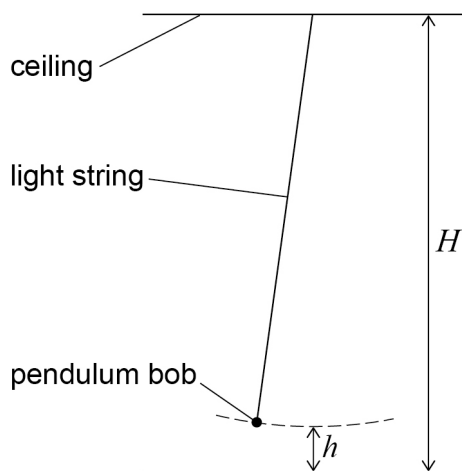
0 9

Figure 8 shows a simple pendulum suspended from a ceiling that is too high to reach.

The height of the ceiling is H .

The centre of mass of the pendulum bob is a distance h from the floor when the pendulum is in the rest position.

Figure 8



A student uses the pendulum to determine H .

The relationship between h and the period T of the pendulum is:

$$T = 2\pi\sqrt{\frac{H-h}{g}}$$

The student wants to use a graphical method to determine both H and g .

0 9 . 1

Describe the procedure that the student should use to collect data that produce accurate determinations of H and g .

[3 marks]



0 9 . 2

Describe how the data can be analysed using a straight-line graph to determine H and g .

[3 marks]

6

Turn over for the next question

Turn over ►



Earthquakes produce seismic waves that travel through the ground and make it vibrate.

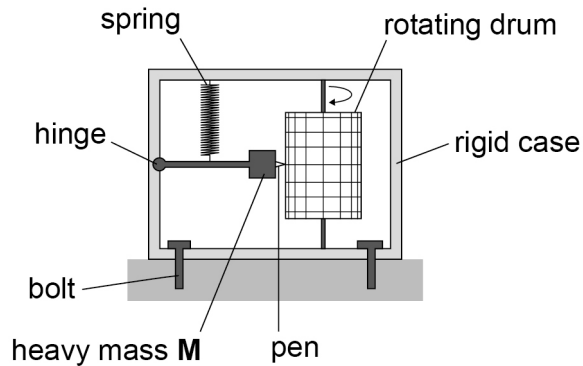
Figure 9 shows a seismograph S_1 that is used to record these vibrations.

A heavy mass M is attached to a rigid case by a frictionless hinge.

M is supported by a spring that has a low stiffness.

The case is bolted to the ground.

Figure 9



During an earthquake the ground surface moves up and down but the centre of mass of M stays almost stationary.

Figure 10 shows S_1 bolted to the ground before the earthquake, with M at its equilibrium position.

Figures 11 and 12 show S_1 during the earthquake.

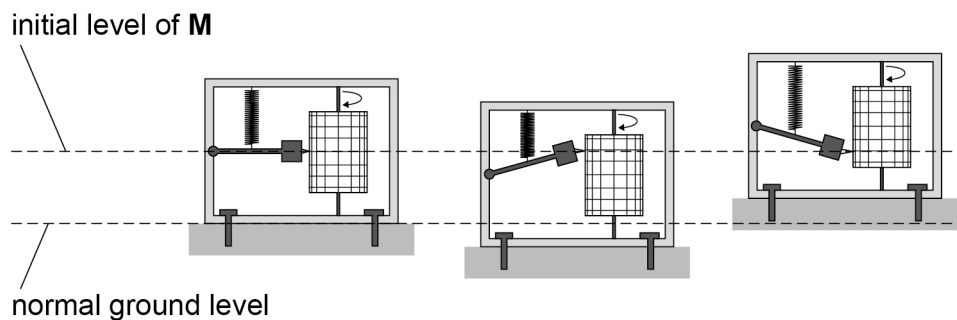
In **Figure 11** the ground has moved down but M has stayed in its initial position.

In **Figure 12** the ground has moved up and M remains in its initial position.

Figure 10

Figure 11

Figure 12

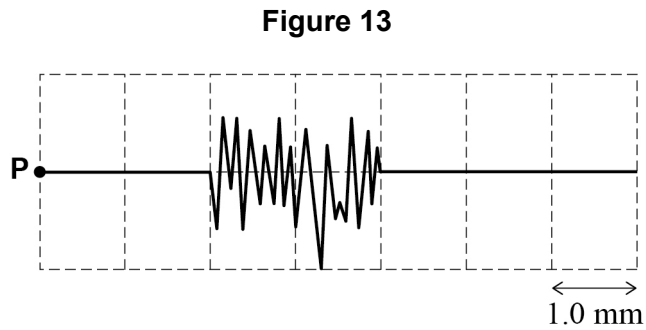


The pen records the movement of the seismograph on graph paper attached to a rotating drum.

1 0 . 1 Explain why **M** stays almost stationary as a seismic wave passes **S₁**.

[1 mark]

Figure 13 shows a magnified image of a trace from **S₁**.
At time $t = 0$ the pen is at position **P**.



The drum has a circumference of 500 mm and a period of rotation of 1000 s.
Each square on the graph paper is 1.0 mm wide.

1 0 . 2 Estimate the frequency of the seismic waves recorded by **S₁**.

[3 marks]

frequency = _____ Hz

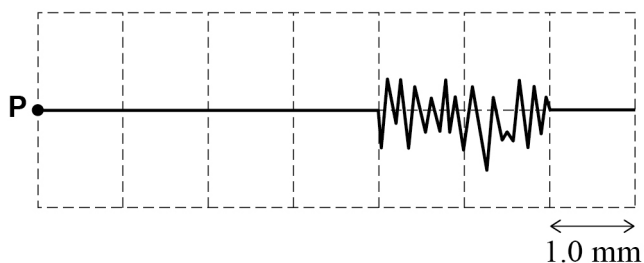
Question 10 continues on the next page

Turn over ►



Figure 14 shows a magnified image of the trace from an identical seismograph **S₂**.

Figure 14



At time $t = 0$ the pen is at position **P**.

S₂ is further from the origin of the earthquake than **S₁**.

The distance between **S₁** and **S₂** is d .

1	0
---	---

.

3

 The speed of the seismic waves is 7.0 km s^{-1} .

Deduce d .

[3 marks]

$d =$ _____ m



1 0 . 4

The time at which the trace starts is different in **Figure 14** compared with **Figure 13**.

Explain **one** other way in which **Figure 14** demonstrates that **S₂** is further than **S₁** from the origin of the earthquake.

[2 marks]

9

END OF SECTION A



Section B

Answer **all** questions in this section.

1 1

Two students do an experiment to determine the resistance per unit length of a metal wire.

Student **A** uses a metre ruler to make a single measurement of the length of the wire. She records a value of 625 mm.

She measures the resistance of the wire as $5.3 \pm 0.3 \, \Omega$.

1 1 . 1

Show that the percentage uncertainty in the length of the wire is approximately 0.2%.

[1 mark]

1 1 . 2

Determine, in $\Omega \, \text{m}^{-1}$, the student's value for resistance per unit length and the absolute uncertainty in her calculated value.

[3 marks]

resistance per unit length = _____ $\Omega \, \text{m}^{-1}$

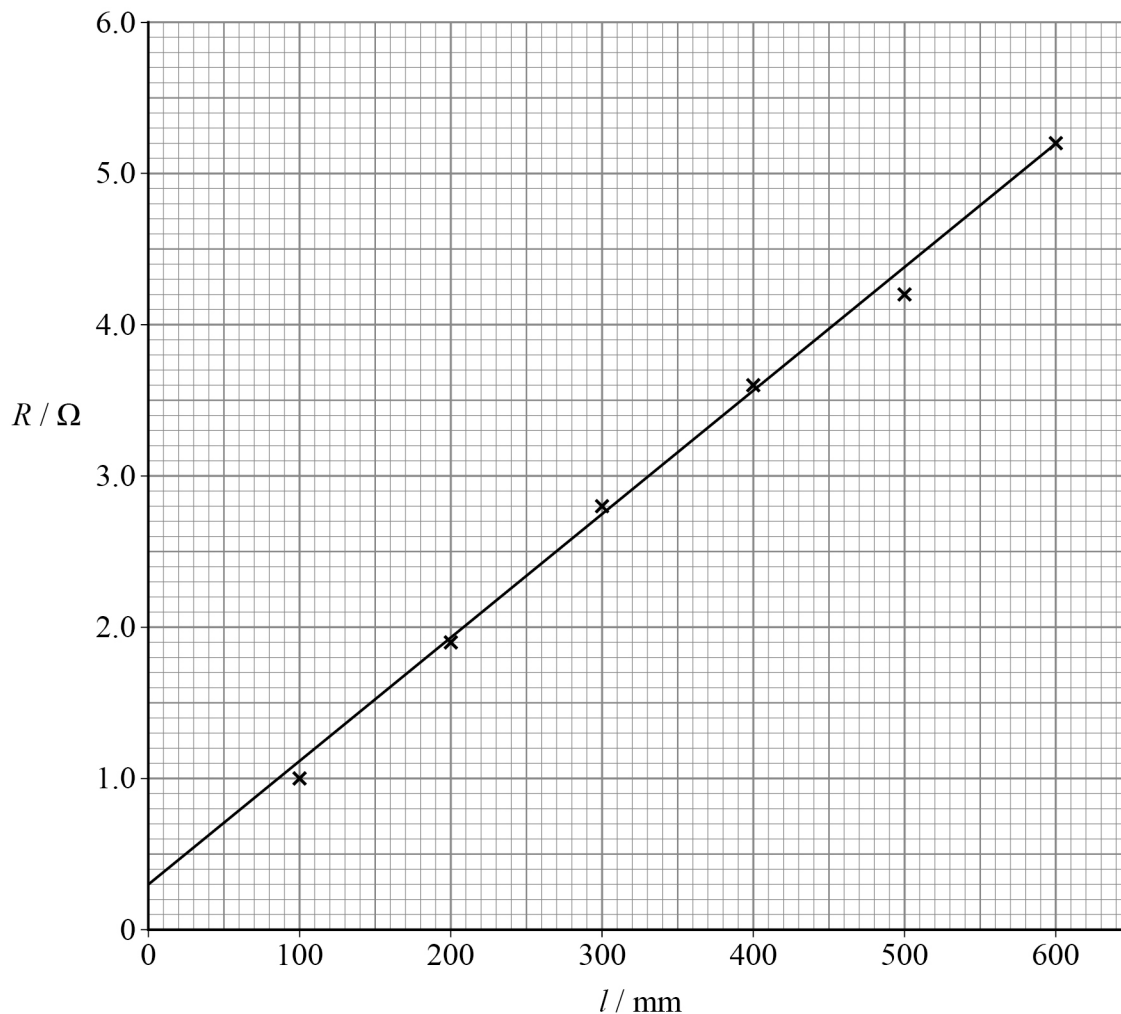
absolute uncertainty = _____ $\Omega \, \text{m}^{-1}$



Student **B** determines values of resistance R for a range of lengths l for the wire and uses them to plot a graph.

Figure 15 shows the graph of R against l .

Figure 15



1 1 . 3

Determine, using the gradient of the graph, the resistance per unit length of the wire in $\Omega \text{ m}^{-1}$.

[2 marks]

resistance per unit length = _____ $\Omega \text{ m}^{-1}$

Question 11 continues on the next page

Turn over ►



1 1 . 4

The method used by student **B** is better than the method used by student **A**.

Suggest **two** reasons why.

[2 marks]

1 _____

2 _____



1 2

A heating element in an electric heater consists of a coil of wire.
The heating element transfers a power of 1.00 kW when connected directly to a mains supply of 230 V.

1 2 . 1

Show that the resistance of the element is approximately 53 Ω .

[2 marks]

1 2 . 2

The element is made from wire that has a radius of 0.137 mm and a resistivity of $4.9 \times 10^{-7} \Omega \text{ m}$.

Calculate the length of wire needed for the element.

[2 marks]

length of wire = _____ m

Question 12 continues on the next page

Turn over ►



1 2 . 3

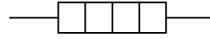
The electric heater contains two elements, each of resistance $53\ \Omega$.
There is also a $230\ \text{V}$ lamp to show when the heater is working.
The power transferred by the lamp is negligible.
The resistivity of the wire does not vary with temperature.

Figure 16 shows the circuit symbols for an ac (mains) supply and an element.

Figure 16



ac (mains) supply



heating element

Draw circuit diagrams to show how the components can be connected to transfer:

- the **maximum** possible power using both elements
- the **minimum** possible power using both elements.

State, for **each** circuit, the total power transferred by the heater.

[4 marks]

maximum

total power transferred = _____ W

minimum

total power transferred = _____ W

END OF SECTION B

Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

1 3

The frequency of the second harmonic of a stationary wave on a string is 240 Hz.
The string is fixed at both ends. The tension and length of the string are kept constant.

What is the frequency of the fifth harmonic of the stationary wave?

[1 mark]

A 96 Hz

☐

B 480 Hz

☐

C 600 Hz

☐

D 1200 Hz

☐

Turn over for the next question

Turn over ►



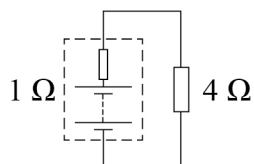
1 4

Each diagram shows a battery connected to an external resistor.
Each battery has an emf of 10 V.
The resistance of each external resistor and the internal resistance of each battery are shown.

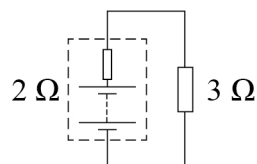
Which circuit transfers the **least** power to the external resistor?

[1 mark]

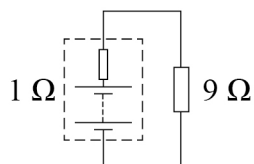
A



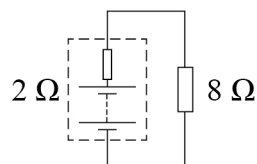
B



C



D



A

☐

B

☐

C

☐

D

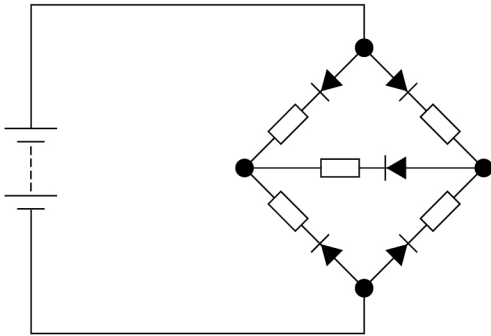
☐


1 5 The diagrams show a battery connected to networks of ideal diodes and resistors.

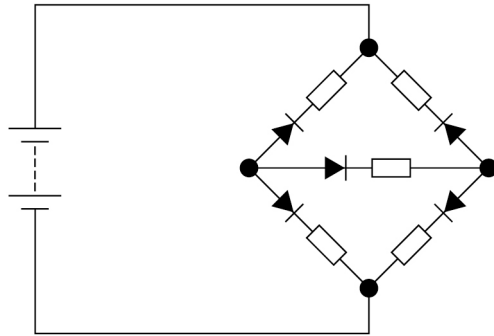
In which circuit will a charge flow in the battery?

[1 mark]

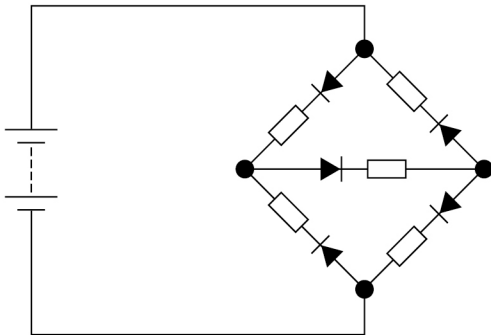
A



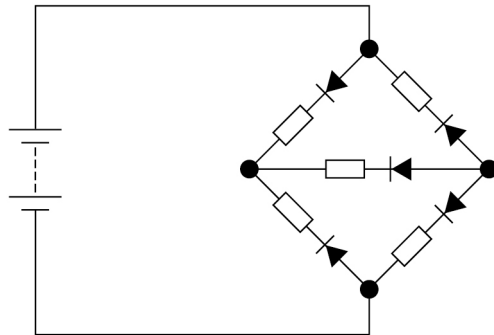
B



C



D



A ☐

B ☐

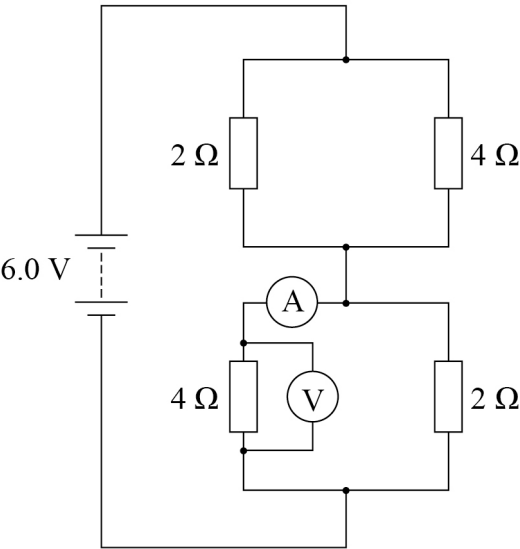
C ☐

D ☐



1 6

A student uses multimeters to measure the potential difference and the current in a circuit. The battery has an emf of 6.0 V and negligible internal resistance.



Which multimeter ranges are most appropriate?

[1 mark]

	Range for voltmeter / V	Range for ammeter / A
A	0–5	0–1
B	0–5	0–5
C	0–10	0–1
D	0–10	0–5

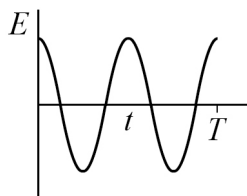
☐
☐
☐
☐


1 7 A body undergoes simple harmonic motion. The period of the motion is T .

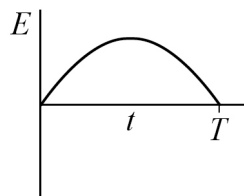
Which graph shows the variation of total energy E with time t ?

[1 mark]

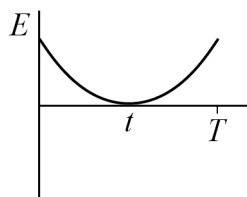
A



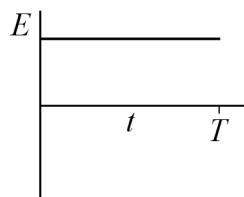
B



C



D



A

☐

B

☐

C

☐

D

☐

Turn over for the next question

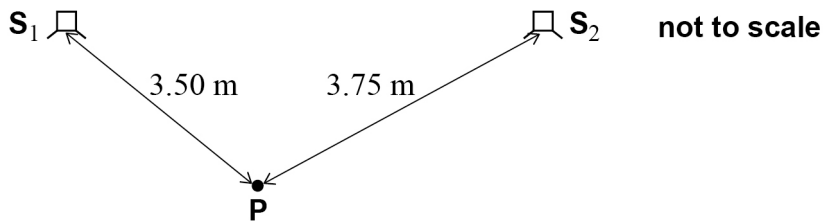
Turn over ►



Questions **18** and **19** refer to two loudspeakers **S₁** and **S₂** separated by a distance of 6.00 m.

1 8

Point **P** is 3.50 m from **S₁** and 3.75 m from **S₂**.



The speed of sound in air is 330 m s^{-1} .

The loudspeakers are in phase and emit sound of frequency 660 Hz.

The phase difference between the waves arriving at **P** from **S₁** and **S₂** is:

[1 mark]

A zero

☐

B $\frac{\pi}{4}$

☐

C $\frac{\pi}{2}$

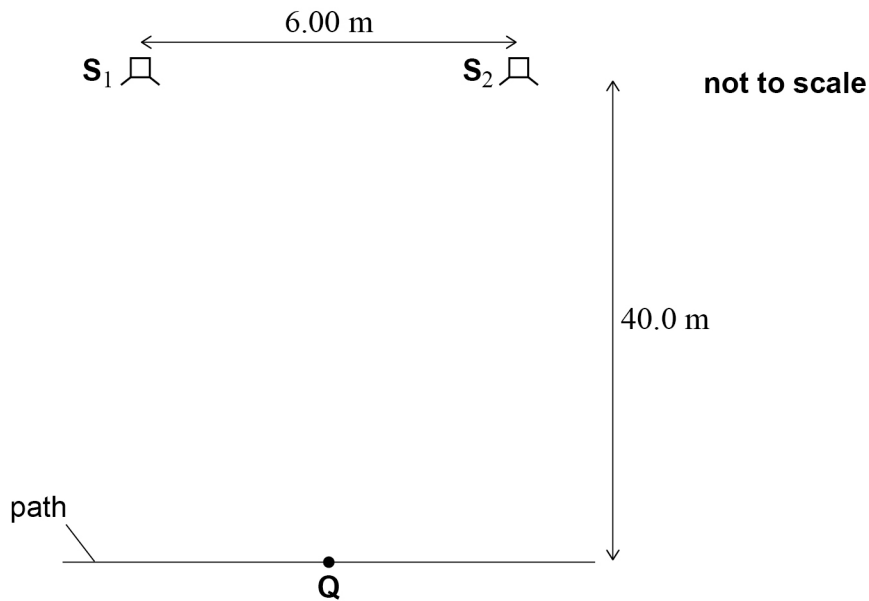
☐

D π

☐


1 9

A path is parallel to the line joining S_1 and S_2 and is 40.0 m from that line.



The sound from both loudspeakers now has a wavelength of 0.70 m.

S_1 and S_2 now emit waves that are π rad out of phase.

The shortest distance along the path between a position with constructive interference and a position with destructive interference is x .

Point Q is equidistant from S_1 and S_2 .

Which row gives x and the amplitude of the sound that is detected at Q ?

[1 mark]

	x / m	Amplitude at Q	
A	2.3	maximum	☐
B	2.3	minimum	☐
C	4.7	maximum	☐
D	4.7	minimum	☐



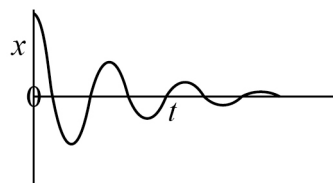
2 0

Damping is applied to an oscillating body at time $t = 0$

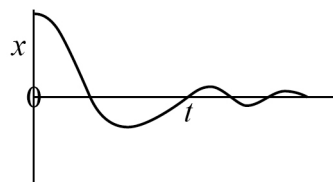
Which graph shows the variation of displacement x of the oscillating body with t ?

[1 mark]

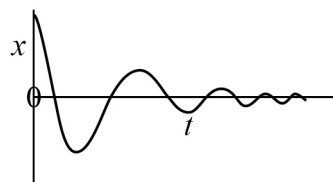
A



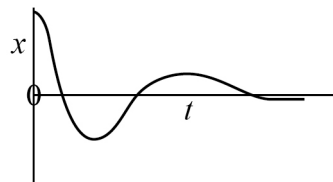
B



C



D

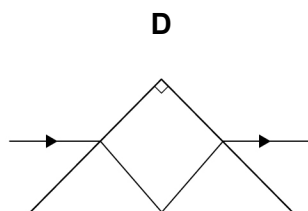
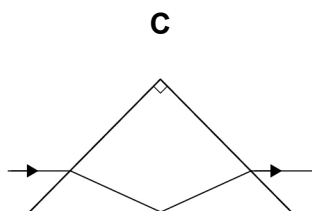
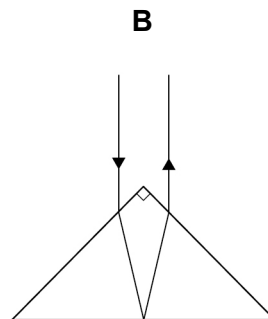
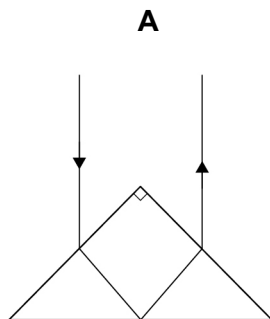


2 1

Monochromatic light passes through a right-angled prism. The critical angle for the glass from which the prism is made is 45° .

Which light path is possible?

[1 mark]



A ☐

B ☐

C ☐

D ☐

2 2

Which statement about dispersion in optical fibres is **not** correct?

[1 mark]

A Modal dispersion is caused by light entering the fibre at different angles. ☐

B Material dispersion is minimised by using a narrow fibre. ☐

C Both material and modal dispersions cause pulse broadening. ☐

D Pulse broadening can be reduced by cladding the fibre. ☐



2 3

Crystal structures are often investigated using electron diffraction.

A student suggests that positron diffraction could be used instead of electron diffraction.

A positron is a positively charged particle that has the same mass and magnitude of charge as an electron.

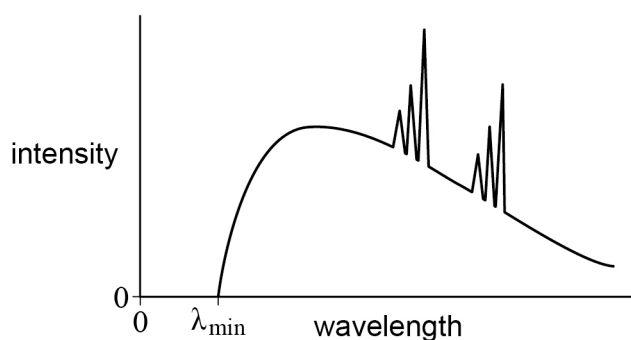
Which statement is correct?

[1 mark]

- A** Electrons have a greater de Broglie wavelength than positrons that have the same energy.
- B** Electrons diffract through a larger angle than positrons that have the same momentum.
- C** Positrons are not diffracted by planes of atoms because of their positive charge.
- D** Positrons exhibit the same wave-like properties as electrons that have the same velocity.

☐☐☐☐

2 **4** The diagram shows the spectrum for an X-ray tube.



The minimum wavelength $\lambda_{\min}$ of X-rays emitted from the X-ray tube can be reduced by

[1 mark]

A reducing the potential difference across the tube.

☐

B increasing the potential difference across the tube.

☐

C reducing the beam current in the tube.

☐

D increasing the beam current in the tube.

☐

Turn over for the next question

Turn over ►



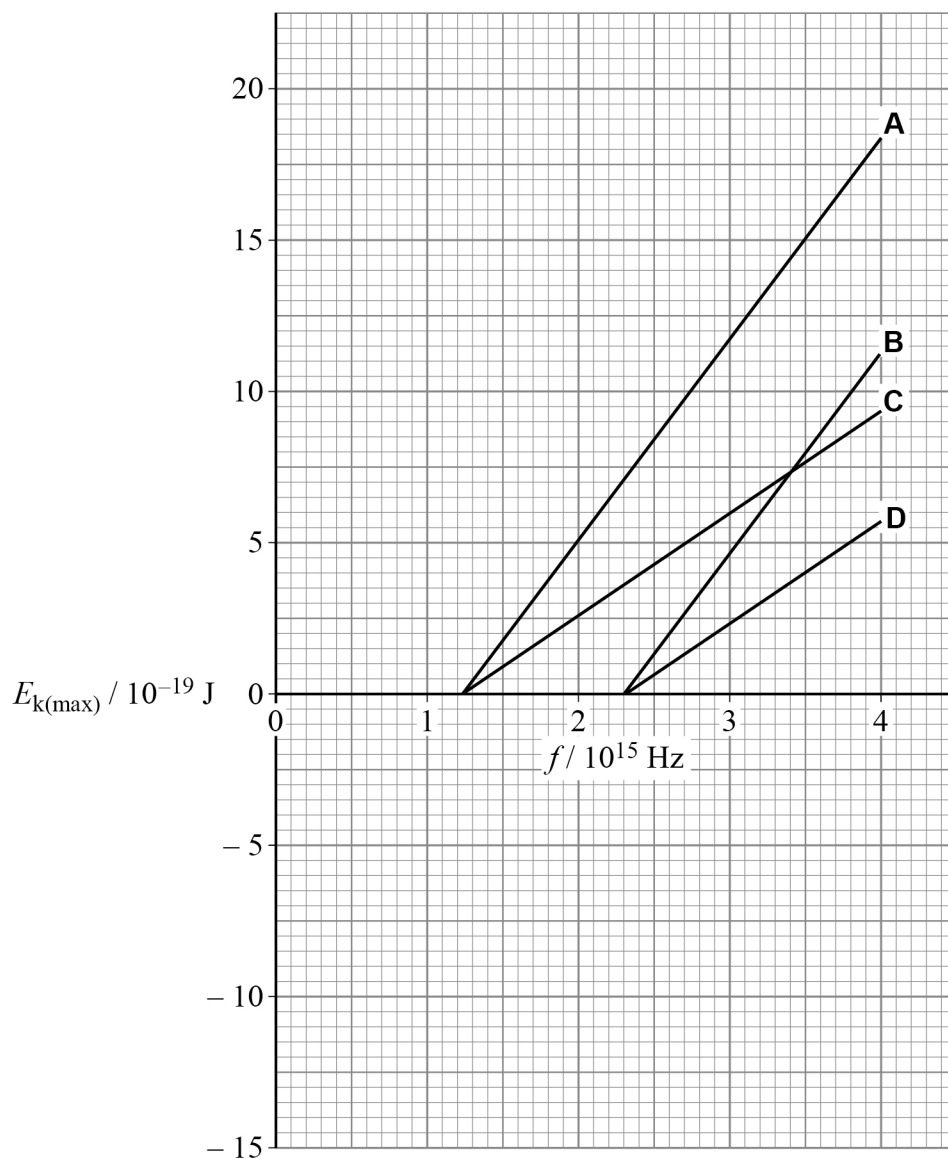
In Questions **25** and **26**, a metal surface is illuminated with radiation of frequency f and photoelectrons are produced.

2 5 The maximum kinetic energy $E_{k(\max)}$ of the photoelectrons is measured for a range of values of f .

The work function of the metal used is $8.2 \times 10^{-19} \text{ J}$.

Which graph shows the variation of $E_{k(\max)}$ with f ?

[1 mark]



- A** ☐
- B** ☐
- C** ☐
- D** ☐



2

6

Which **single** change will cause the stopping potential of the photoelectrons to increase?

[1 mark]

A increasing the number of photons per second that are incident on the surface ☐

B increasing the wavelength of the incident radiation ☐

C increasing the frequency of the incident radiation ☐

D using a metal of greater work function ☐

*Do not write
outside the
box*

14

[illegible]

[illegible]

[illegible]

0 8

Two loudspeakers emit sound waves that are coherent.

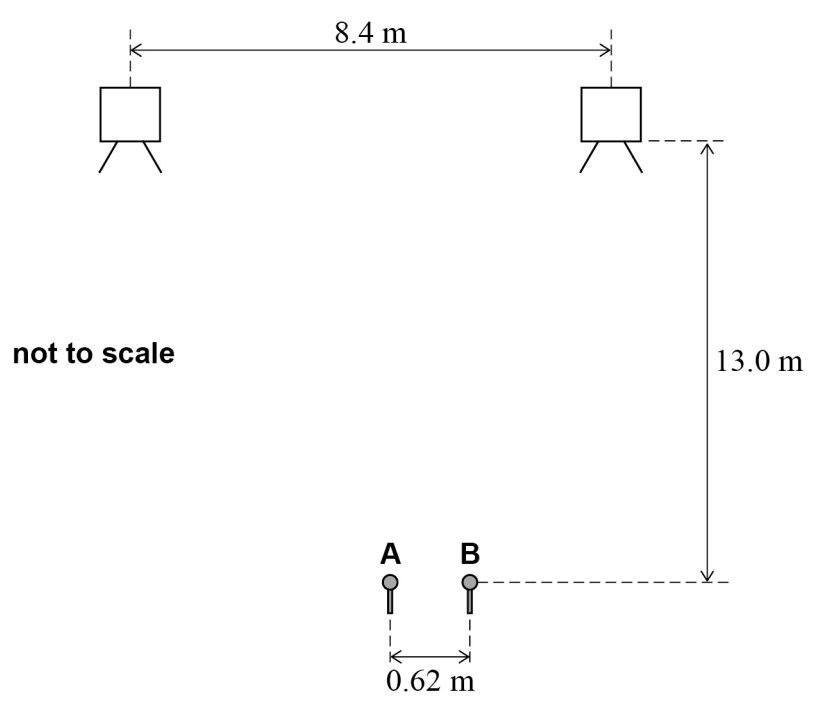
0 8 . 1

State what is meant by coherent.

[2 marks]

The two loudspeakers are 8.4 m apart in a large open space, as shown in **Figure 7**. Two microphones **A** and **B** are placed 0.62 m apart in front of the loudspeakers. **A** is the same distance from each speaker. The perpendicular distance between the loudspeakers and the microphones is 13.0 m.

Figure 7



Microphone **A** detects a sound signal from the loudspeakers. The sound intensity detected at microphone **B** is zero.



0 8 . 2

Explain why the intensity of sound at microphone **B** is zero.

[3 marks]

0 8 . 3

Calculate the frequency of the sound emitted by the loudspeakers.
There is no other position of zero intensity between **A** and **B**.
The speed of sound in air is 340 m s^{-1} .

[3 marks]

frequency = _____ Hz

Question 8 continues on the next page

Turn over ►



0 8 . 4

The frequency of the sound from the loudspeakers is slowly increased.
The loudspeakers continue to be coherent sources.

Explain how this frequency increase affects the intensity of sound detected by
microphones **A** and **B**.

[2 marks]

0 8 . 5

The arrangement of loudspeakers and microphones is placed inside a room.

Suggest why the maximum and minimum intensities of sound detected by **A** and **B**
are not as well defined when the arrangement is placed in the room.

[2 marks]

END OF SECTION A



Section B

Answer **all** questions in this section.

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outside the
box*

1

5

A vertical spring of stiffness k supports an object of mass m .

The frequency of oscillation of the mass–spring system is f .

A second vertical spring has a stiffness $2k$ and supports an object of mass $3m$.

What is the frequency of oscillation of the second mass–spring system?

[1 mark]

A $1.5f$

☐

B $1.2f$

☐

C $0.82f$

☐

D $0.67f$

☐

Ultrasound is used in medical diagnosis because high-frequency sound waves

[1 mark]

A change frequency when they cross tissue boundaries.

☐

B are not attenuated in soft tissues.

☐

C travel at the same speed in different tissues.

☐

D are reflected well at tissue boundaries.

☐

Turn over ►



Do not write

A stretched string has a length l and a tension T .

The mass per unit length of the string is μ .

A second string has a length $0.6l$ and mass per unit length 0.8μ .

What is the tension of the second string when it vibrates with the same fundamental frequency as the first string?

[1 mark]

A $0.23T$

☐

B $0.29T$

☐

C $0.45T$

☐

D $0.48T$

☐

1**9**

Loudspeaker **P** emits sound of frequency 50 Hz.
Loudspeaker **Q** emits sound of frequency 60 Hz.
At time $t = 0$ **P** and **Q** are in phase.

When will **P** and **Q** next be in phase?

[1 mark]

A $t = 0.10$ s

☐

B $t = 0.25$ s

☐

C $t = 0.50$ s

☐

D $t = 0.75$ s

☐

0	1
---	---

Describe the nature of transverse waves.

[2 marks]

2

0 2

Superconductors can be used in the electromagnets in some types of train.

0 2 . 1

Explain what is meant by a superconductor.

[2 marks]

0 2 . 2

Explain why superconductors are used in these electromagnets.

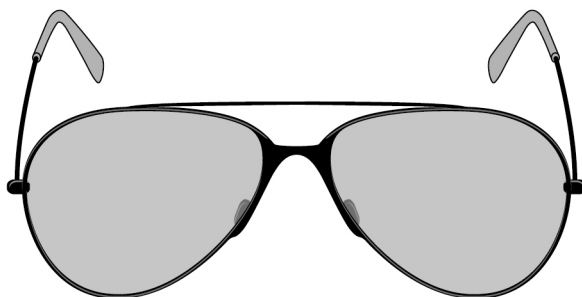
[2 marks]



Don't write

Figure 1 shows sunglasses that have polarising lenses. The sunglasses are sold with a separate small polarising filter.

Figure 1



polarising
filter

State and explain how the filter can be used to check that the lenses are polarising. **[3 marks]**

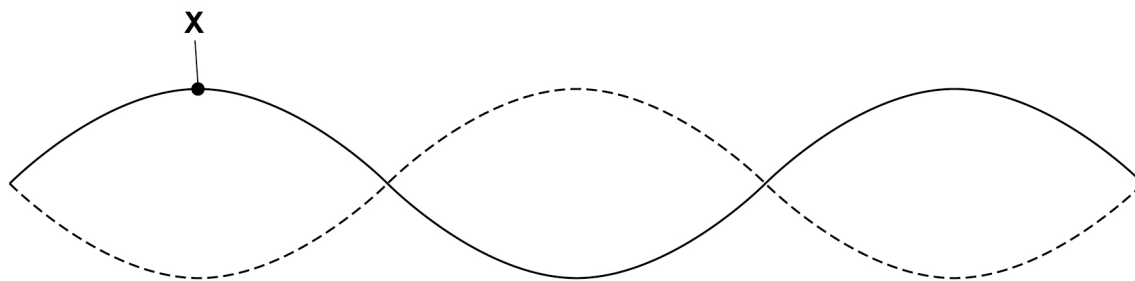


Do not write Do not write

0 5

Figure 2 represents a stationary wave. **X** marks the position of an antinode.

Figure 2



0 5

1

Explain how the antinode is produced.

[2 marks]

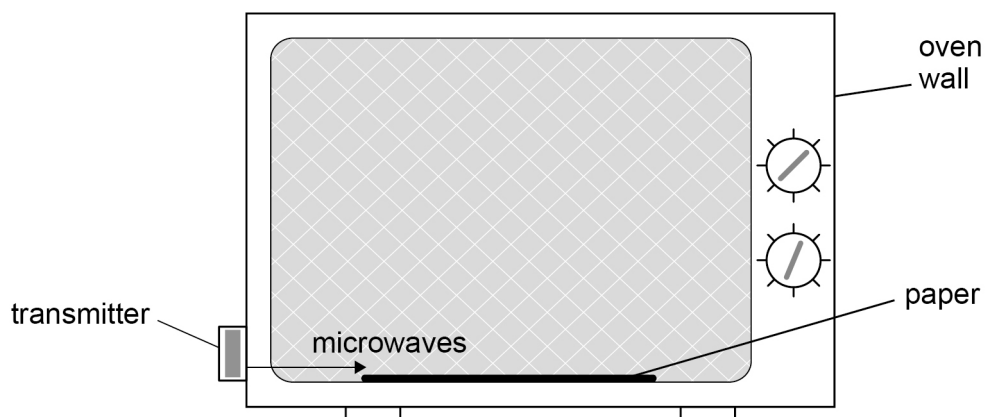
0 5

2

A student determines the speed of microwaves in a microwave oven from which the turntable has been removed. Stationary waves are formed in the oven.

The student has paper that changes colour above a certain temperature. The paper is placed on the floor of the oven as shown in **Figure 3**.

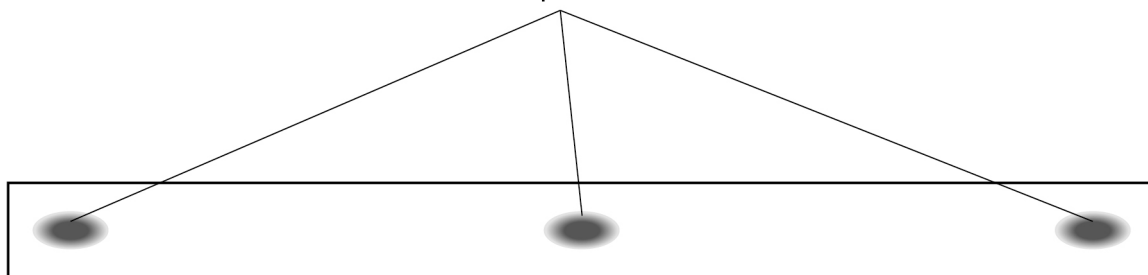
Figure 3



The student switches on the oven for a short time and then removes the paper.
Figure 4 shows the paper at actual size. There are three spots where the paper has changed colour.

Figure 4

spots



The frequency of the emitted microwaves is 2.45 GHz.

Determine the speed of the microwaves in the oven by taking measurements from **Figure 4**.

[3 marks]

speed = _____ m s⁻¹



Do not write Do not write

1**2**

Light of wavelength 480 nm travels as a pulse in air.
The duration of the pulse is 10 μs .

How many complete waves does the pulse contain?

[1 mark]

A 6.25×10^9

☐

B 6.25×10^{12}

☐

C 6.25×10^{14}

☐

D 6.25×10^{19}

☐

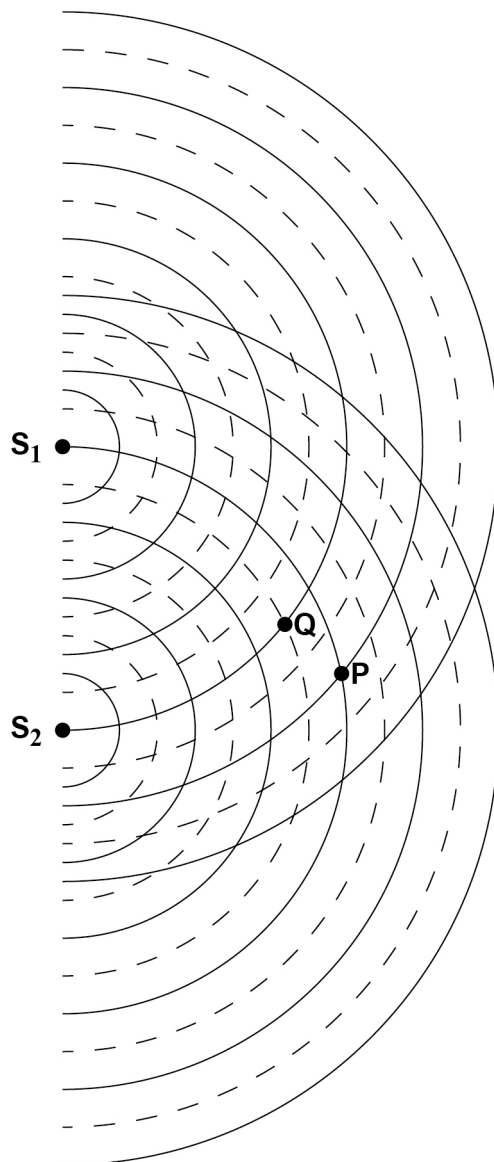
Turn over for the next question

Turn over ►



Do not write Do not write

The diagram shows wave fronts from two sources S_1 and S_2 .
The peaks of the waves are represented by dashed lines ----- .
The troughs of the waves are represented by continuous lines ——— .



P and **Q** show two points where the waves superpose.

Which is correct?

[1 mark]

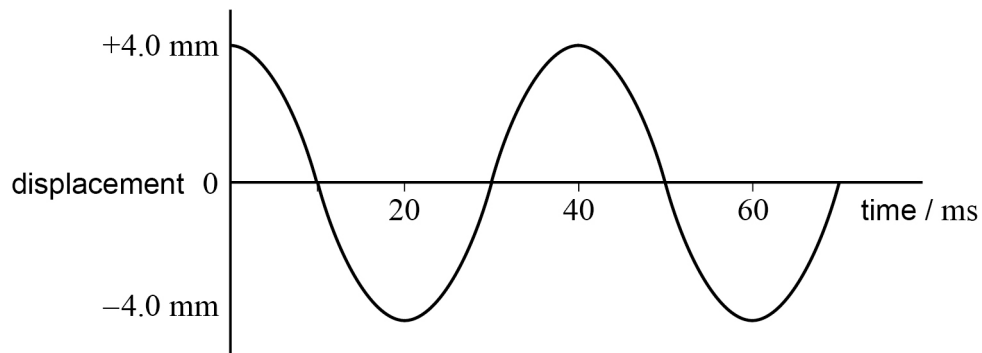
- A** The waves arrive in phase at **Q**. ☐
- B** The waves interfere destructively at **P**. ☐
- C** The amplitude is a minimum at **Q**. ☐
- D** The path difference is an odd number of wavelengths at **Q**. ☐



Do not write	Do not write
--------------	--------------

2 3

The diagram shows how the displacement of a particle in a progressive wave varies with time.



Which row gives the amplitude and frequency of the wave?

[1 mark]

	Amplitude / mm	Frequency / Hz	
A	4.0	25	☐
B	8.0	25	☐
C	4.0	40	☐
D	8.0	40	☐

0	4
---	---

A communications system sends information through an optical fibre as pulses of light. Pulse broadening limits the rate at which the information can be transmitted in the system.

0	4	.	1
---	---	---	---

Explain how material dispersion causes pulse broadening in the optical fibre.

[2 marks]

0	4	.	2
---	---	---	---

Explain why the maximum rate at which information can be transmitted depends on the length of the optical fibre.

[2 marks]



0 5 . 1 Ultrasound waves are used for imaging in hospitals.

State **one** other medical application of ultrasound.

[1 mark]

0 5 . 2 One ultrasound imager emits waves of frequency 3.2 MHz.

Table 1 shows the speeds of the ultrasound waves as they pass through three types of tissue in a human body.

Table 1

Body tissue	Speed of wave / km s ⁻¹
fat	1.4
muscle	1.6
bone	3.5

Determine the maximum wavelength of the ultrasound as it passes through the body.

[2 marks]

maximum wavelength = _____ m

3



0 | 6

The frequency f of the first harmonic of a vibrating string depends on the variables T , l and μ as shown by the equation:

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

06

1

State the meaning of μ in this equation.

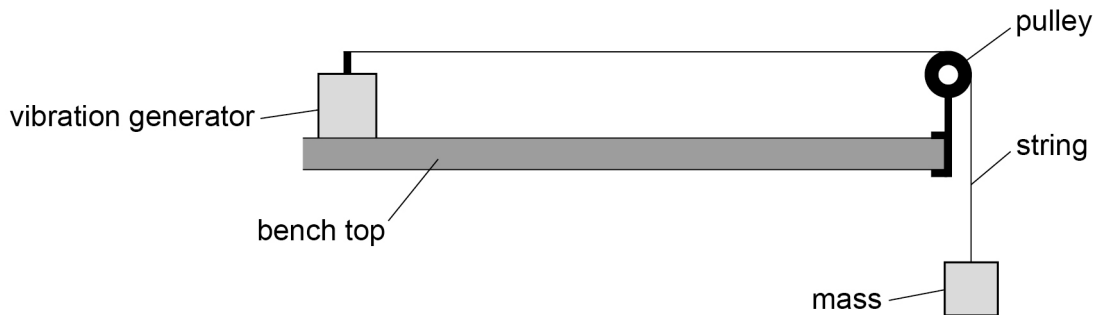
[1 mark]

0 | 6

2

Figure 2 shows apparatus used to produce stationary waves on a stretched string.

Figure 2



Describe how this apparatus can be used to collect the data needed to investigate the variation of f with l .

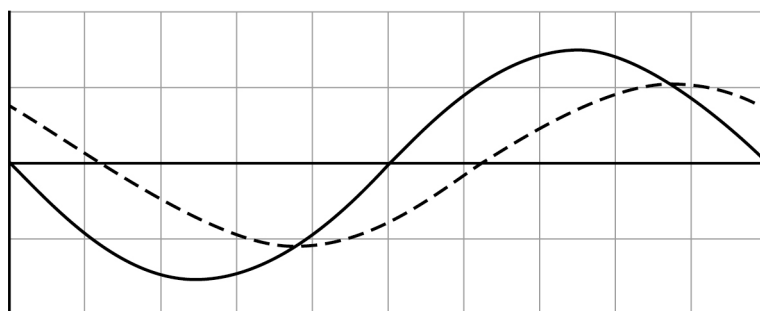
[4 marks]

[illegible]

Figure 3 shows a stationary wave **W** on the full length of a vibrating string at a particular time t . The stationary wave **W** can be described as the superposition of two progressive waves **X** and **Y**.

X and **Y** have the same frequency and travel in opposite directions.
The dashed line on **Figure 3** shows the position of **X** at time t .

Figure 3



Key

—— stationary wave **W**
---- progressive wave **X**

0 6 . 3 State which harmonic of **W** is shown in **Figure 3**.

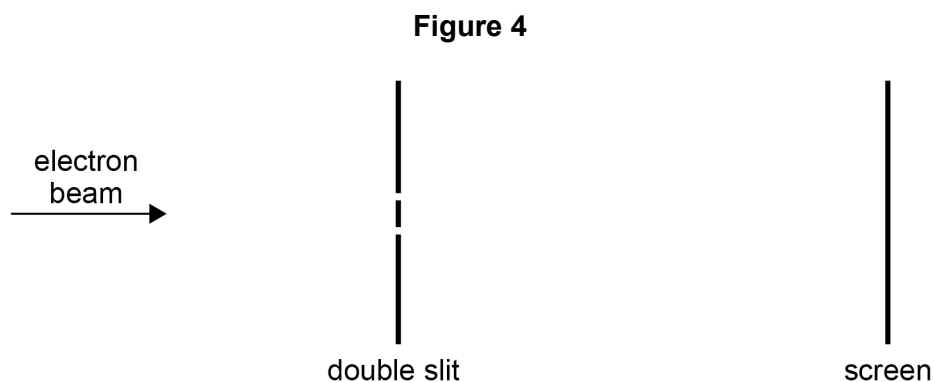
[1 mark]

0 6 . 4 Sketch on **Figure 3** the progressive wave **Y**.

[2 marks]

07

Figure 4 shows a simplified arrangement of apparatus used to investigate the wave behaviour of electrons at a double slit.



07.1

Electrons in the beam have been accelerated from rest across a potential difference V to reach a final speed v .

Show that:

$$v = \sqrt{\frac{2Ve}{m_e}}$$

[1 mark]

07.2

Show that when $V = 50.0 \text{ kV}$ the de Broglie wavelength of the electrons is approximately $5.5 \times 10^{-12} \text{ m}$.

[2 marks]



0 7 . 3

The electrons arrive at the double slit.
The spacing between the two slits is $2.0 \mu\text{m}$.

Calculate the fringe spacing at a distance of 0.35 m from the slits.

[2 marks]

fringe spacing = _____ m

A fringe pattern is observed when electrons arrive at the screen.
Figure 5 shows a magnified image of this pattern.

Figure 5



0 7 . 4

State **two** wave behaviours that explain how the electrons produce this fringe pattern.

[2 marks]

1 _____

2 _____

Question 7 continues on the next page

Turn over ►



0 7 . 5

The accelerating potential difference V is now increased, causing the fringe spacing to change.

Explain the change to the fringe spacing.

[2 marks]

0 7 . 6

Monochromatic light passes through a double slit. Coherent light from the two slits produces a fringe pattern similar to that in **Figure 5**.

Explain why this fringe pattern occurs.

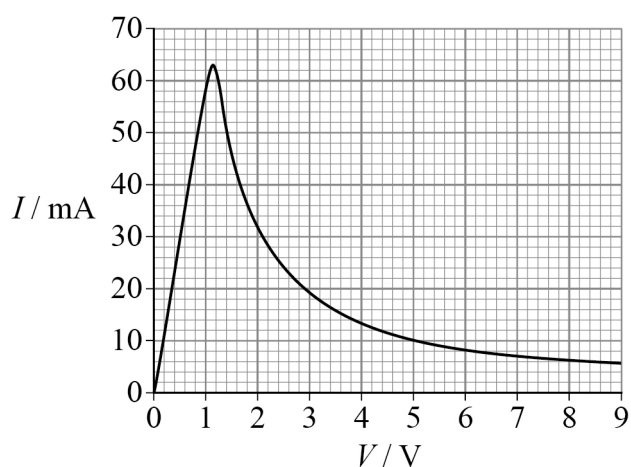
[3 marks]



08

Figure 6 shows the variation of current I with potential difference V for a component as V increases.

Figure 6



08.1

Discuss the extent to which this component obeys Ohm's law.

[2 marks]

Question 8 continues on the next page



Figure 7 shows a thermistor **T** in a circuit. The circuit is used to monitor the temperature of water. The battery has an emf of 10.0 V and negligible internal resistance. The fixed resistor **R** has a resistance of 6.1 k Ω .

Figure 7

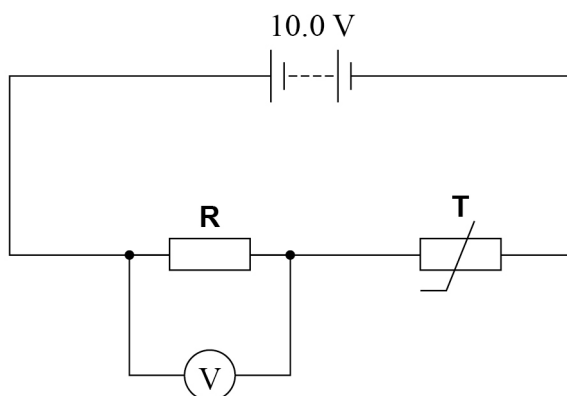
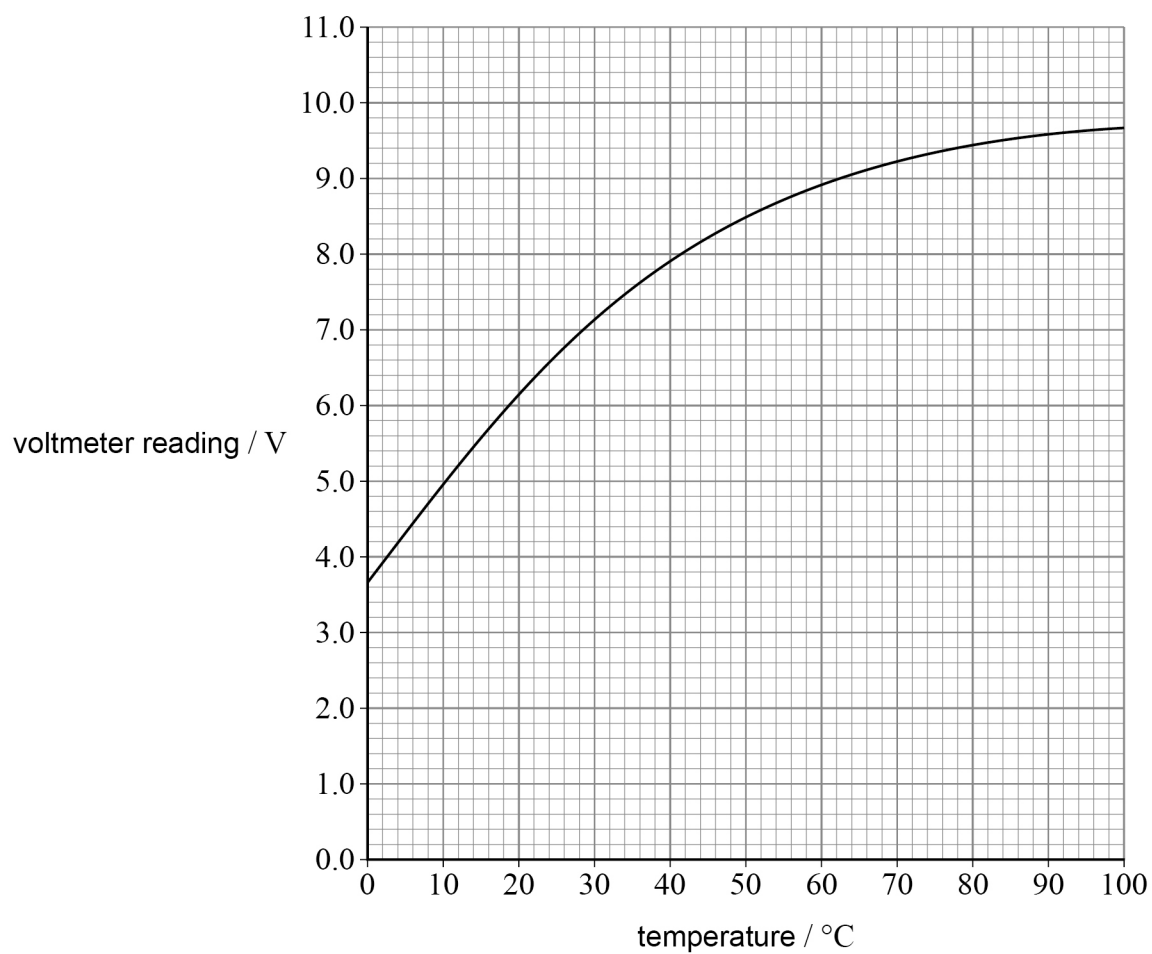


Figure 8 shows how the voltmeter reading varies with the temperature of **T**.

Figure 8



0 8 . 2

Explain why the voltmeter reading changes as the temperature increases.

[2 marks]

0 8 . 3

Show that the resistance of **T** at 55 °C is approximately 900 Ω .

[2 marks]

Question 8 continues on the next page

Turn over ►



Figure 9 shows a modification to the circuit with a second fixed resistor **S** parallel to **T**.

Figure 9

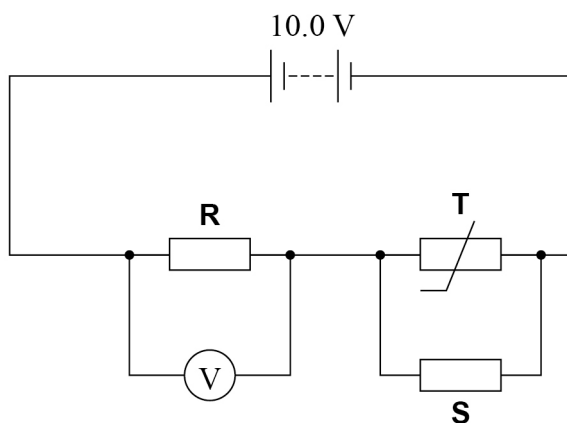
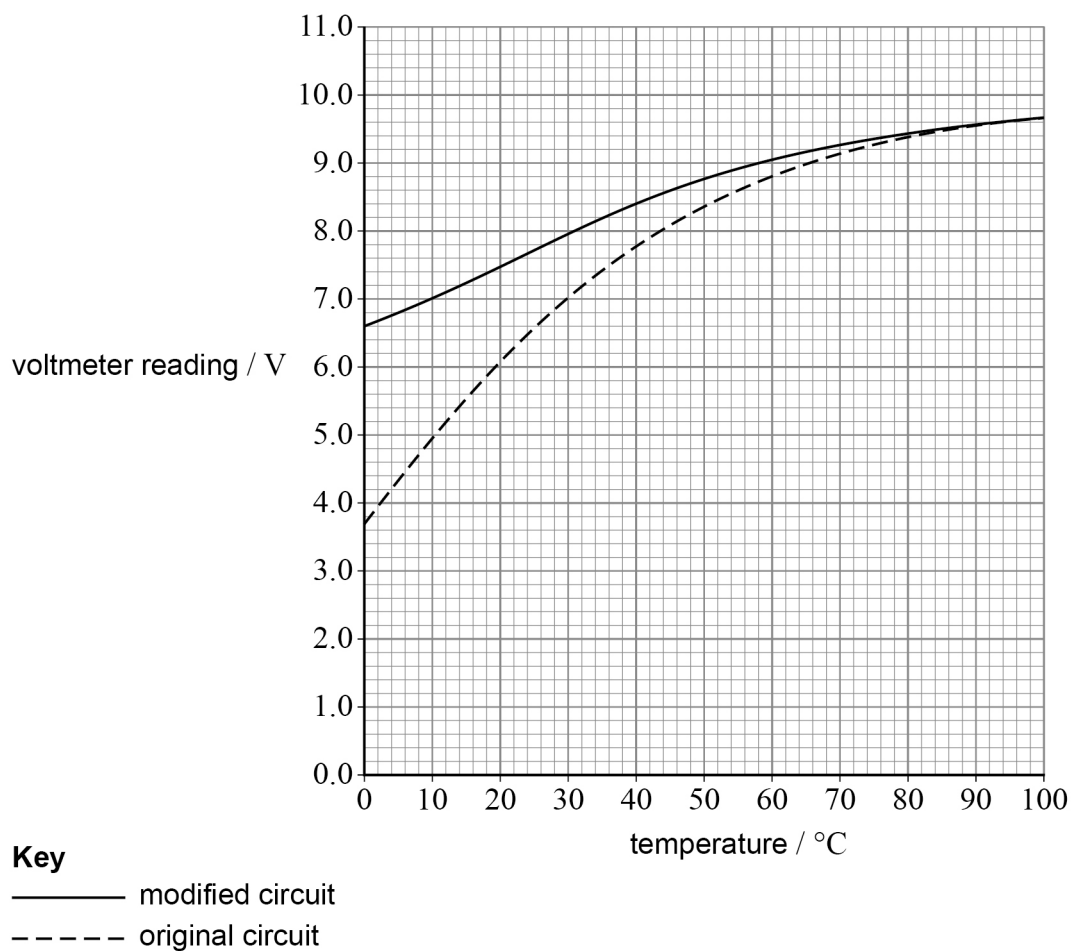


Figure 10 shows how the voltmeter reading now varies with the temperature of **T** for this modified circuit. **Figure 10** also shows the voltage variation with temperature for the original circuit in **Figure 7**.

Figure 10



0 8 . 4

The current in **R** is 1.46 mA when the temperature of **T** is 55 °C.

Determine the resistance of **S**.

[3 marks]

resistance of **S** = _____ Ω

0 8 . 5

Suggest **one** disadvantage of using the modified circuit rather than the original circuit to monitor the temperature of the water between 0 °C and 100 °C.

[1 mark]

10

Turn over for the next question

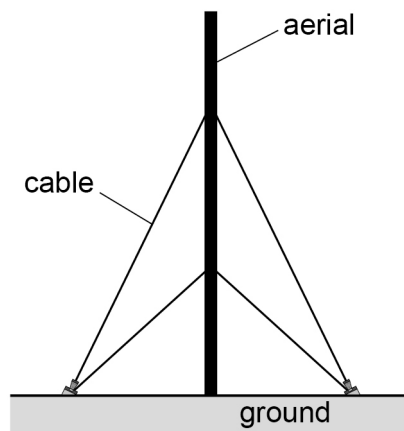
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0 9

Figure 11 shows an aerial fixed in a vertical position by cables. The aerial transmits radio waves.

Figure 11



0 9 . 1

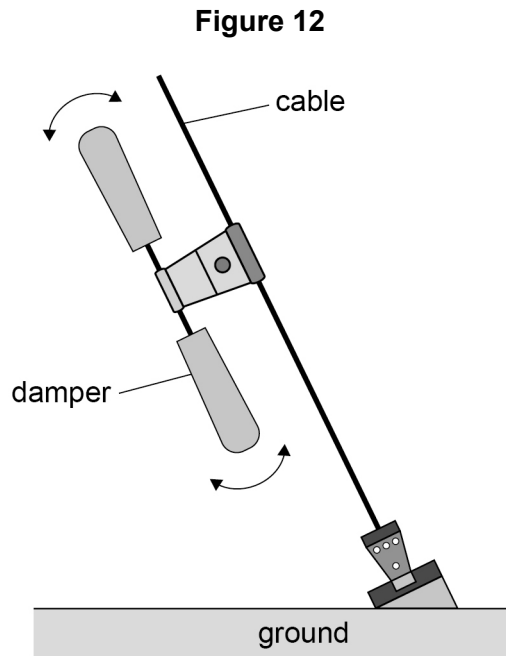
Explain why aerials that receive these radio waves should be vertical.

[2 marks]



The cables vibrate in the wind. The vibrations can damage the cables.
To minimise this damage a damper is attached to each cable.

Figure 12 shows a damper attached to a cable. The ends of the damper are free to oscillate as shown.



0 9 . 2 State what is meant by damping.

[1 mark]

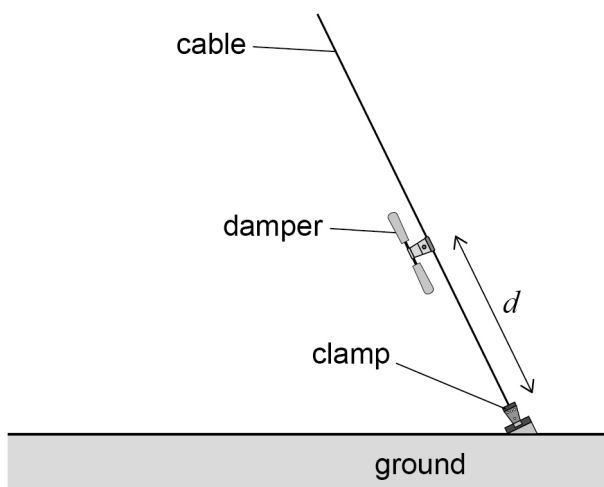
Question 9 continues on the next page



0 9 . 3

The damper is most effective when placed at a particular distance d from the clamp.
Figure 13 shows one damper at d .

Figure 13



Engineers determine d by modelling the harmonics that can form on the cable.

Suggest why d is determined by using the harmonic with the maximum frequency that can form on the cable.

[2 marks]

END OF SECTION A



Turn over for Section B

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ANSWER IN THE SPACES PROVIDED**

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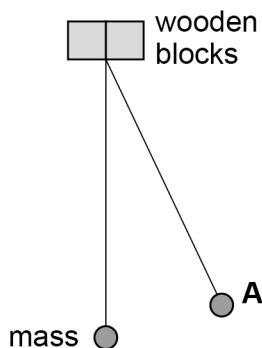


Section B

Answer **all** questions in this section.

1 0

A student uses a simple pendulum to determine a value for the acceleration g due to gravity. He attaches a small mass to a light string suspended between two wooden blocks, as shown in **Figure 14**.

Figure 14**not to scale**

The student uses a metre ruler to measure the length L of the pendulum and records the value as $L = 400 \text{ mm} \pm 1 \text{ mm}$.

1 0 . 1

Explain why the student records the absolute uncertainty in L as $\pm 1 \text{ mm}$.

[1 mark]

He then moves the mass through a small angle to position **A**. He releases the mass and determines the time period T by noting the time T_{10} for 10 oscillations.

He records this as $T_{10} = 12.8 \text{ s} \pm 0.1 \text{ s}$.

1 0 . 2

Show that the percentage uncertainty in T is approximately 0.8%.

[1 mark]

1 0 . 3

Determine the student's value for g .

[2 marks]

$g =$ _____ m s^{-2}

1 0 . 4

Calculate the percentage uncertainty in g .

[2 marks]

percentage uncertainty = _____

1 0 . 5

Suggest **two** ways in which the student can increase the accuracy of his determination of g .

[2 marks]

1 _____

2 _____



Figure 15 shows a fuel gauge that uses an optical fibre to measure the level of fuel in a container. Light passes along the optical fibre to a detector. The fibre has a series of V-grooves in one of its vertical sections.

Figure 15

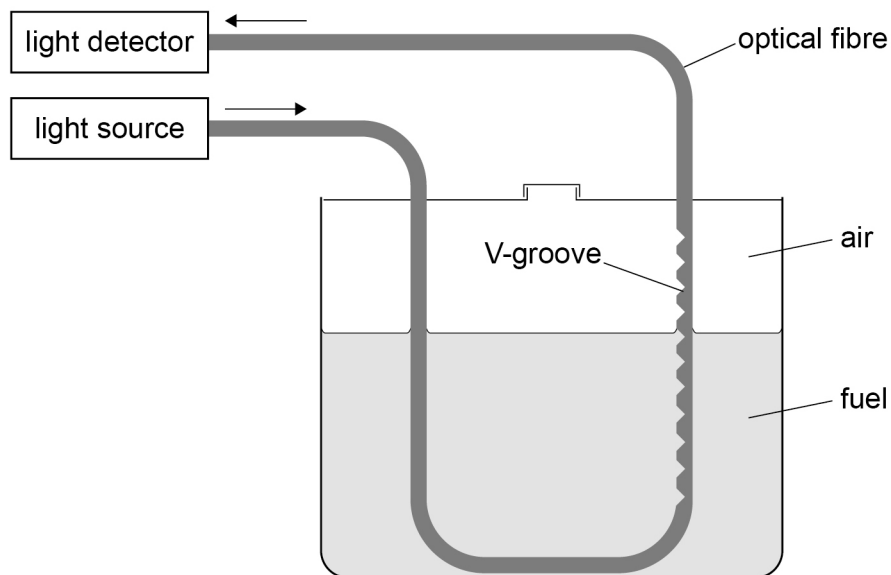
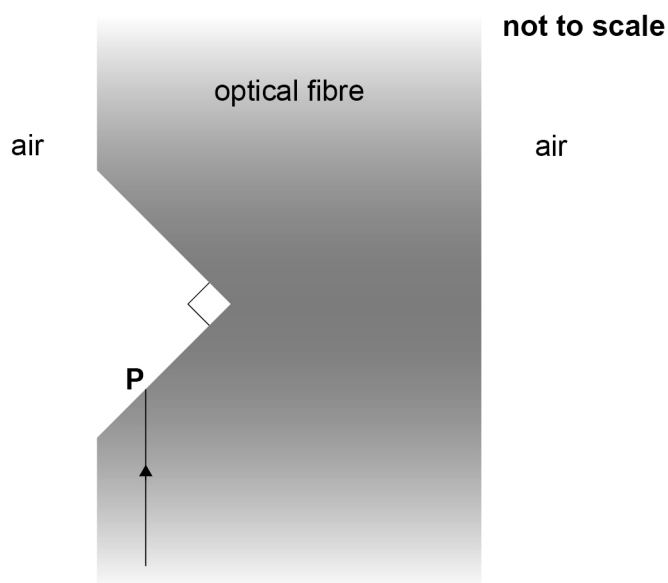


Figure 16 shows a magnified image of one of the grooves when the optical fibre is vertical and surrounded by air. Each groove has an angle of 90° . Light travels vertically up the fibre and is incident on the lower section of the groove at **P**.

Figure 16



1 1 . 1 The refractive index of the optical fibre is 1.49

Show that the critical angle for the optical fibre in air is approximately 42° .

[1 mark]

1 1 . 2 Explain where the light ray shown in **Figure 16** leaves the optical fibre.

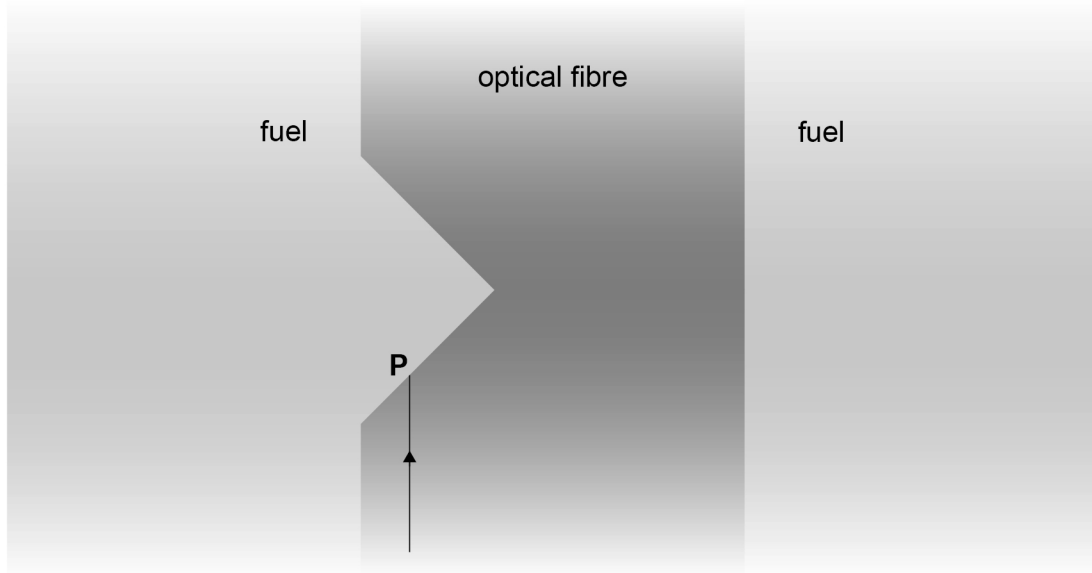
[2 marks]

Question 11 continues on the next page



Figure 17 shows the groove when surrounded by fuel. The refractive index of the fuel is 1.39

Figure 17



not to scale

1 1 . 3

Show that the angle of refraction at **P** is approximately 50° .

[2 marks]

1 1 . 4

Draw on **Figure 17** the path of the ray through the fuel after the ray leaves **P**.

[1 mark]

1 1 . 5

Explain how the light intensity at the detector changes as the level of the fuel rises in the tank.

[2 marks]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

1 2 A cell has an emf of 4.00 V and an internal resistance of 3.00 Ω .

An 18.0 Ω resistor is connected across the cell.

What is the terminal pd?

[1 mark]

A 0.57 V

☐

B 0.67 V

☐

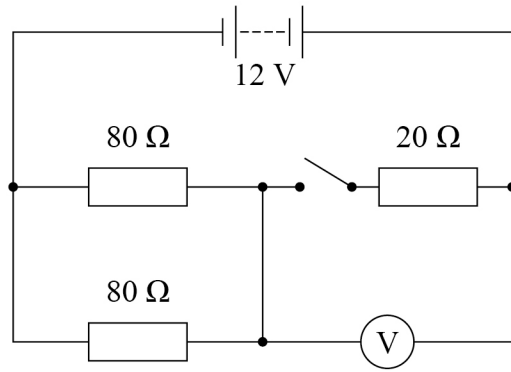
C 3.33 V

☐

D 3.43 V

☐


1 3 The battery in a circuit has an emf of 12 V and negligible internal resistance.



What is the voltmeter reading when the switch is open and when it is closed?

[1 mark]

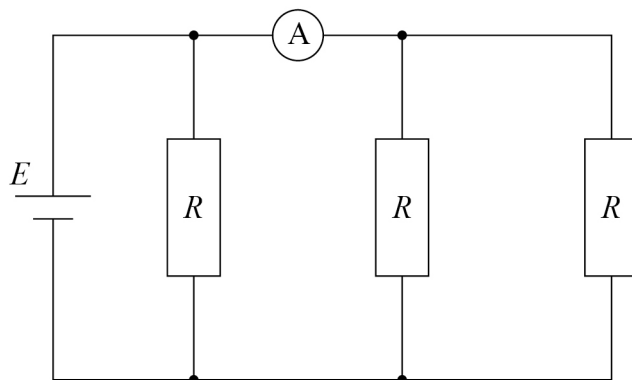
	Switch open	Switch closed	
A	0 V	6 V	☐
B	0 V	4 V	☐
C	12 V	6 V	☐
D	12 V	4 V	☐



1 4

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box

Three resistors and an ammeter are connected to a cell. The cell has an emf E and negligible internal resistance. Each resistor has a resistance R .



What is the reading on the ammeter?

[1 mark]

A $\frac{ER}{3}$

B $\frac{2ER}{3}$

C $\frac{2E}{R}$

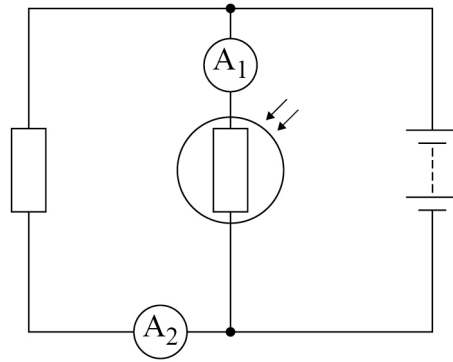
D $\frac{3E}{R}$

Turn over for the next question

Turn over ►



- 1 5** The diagram shows a battery connected to an LDR, a fixed resistor and two ammeters A_1 and A_2 . The internal resistance of the battery is negligible.



The light intensity incident on the LDR increases.

What will be observed on A_1 and A_2 ?

[1 mark]

	A_1	A_2	
A	decreasing value	no change	☐
B	increasing value	no change	☐
C	decreasing value	decreasing value	☐
D	increasing value	decreasing value	☐



1 6 The diagram shows three energy levels for an atom.

The vertical positions of the energy levels are to scale.

level 2 _____
level 1 _____

ground state _____

Electron transitions between these levels produce photons with frequencies of:

$5.62 \times 10^{14} \text{ Hz}$
 $3.78 \times 10^{15} \text{ Hz}$
 $4.34 \times 10^{15} \text{ Hz}$.

What is the energy difference between level 1 and the ground state?

[1 mark]

A $3.7 \times 10^{-19} \text{ J}$ ☐

B $2.1 \times 10^{-18} \text{ J}$ ☐

C $2.5 \times 10^{-18} \text{ J}$ ☐

D $2.9 \times 10^{-18} \text{ J}$ ☐

Turn over for the next question

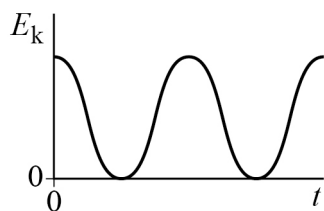
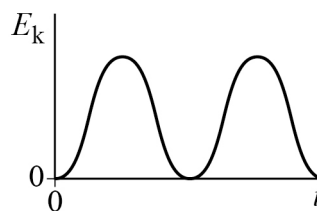
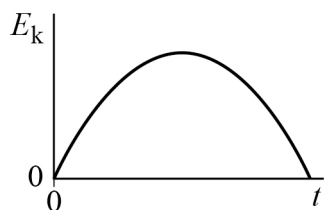
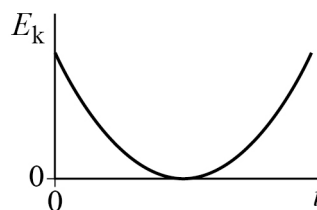
Turn over ►



1 7

A mass–spring system is displaced and then released at time $t = 0$. It oscillates freely.

Which graph shows the variation of kinetic energy E_k with time t for one full oscillation of the system?

[1 mark]**A****B****C****D****A**☐**B**☐**C**☐**D**☐

1 8

Wire **X** is made of a material with a resistivity ρ .
X has a length l , a resistance R and a diameter d .

Wire **Y** is made of a material with a resistivity 4ρ .
Y has a length $0.5l$ and a resistance $4R$.

What is the diameter of **Y**?

[1 mark]

A $\frac{d}{2}$

☐

B $\frac{d}{\sqrt{2}}$

☐

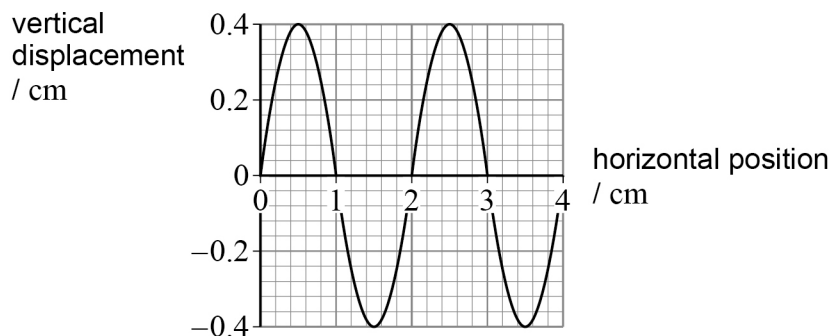
C $\sqrt{2}d$

☐

D $2d$

☐
1 9

A water wave of frequency 10 Hz moves on the surface of a lake. The graph shows the variation of vertical displacement with horizontal position for the water surface at an instant in time.



What is the speed of the water wave?

[1 mark]

A 0.20 cm s^{-1}

☐

B 4.0 cm s^{-1}

☐

C 10 cm s^{-1}

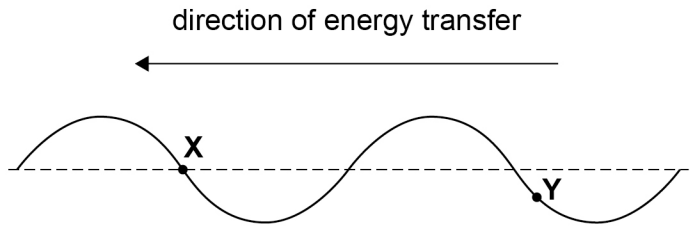
☐

D 20 cm s^{-1}

☐

2 0

A progressive wave moves from right to left on a string.
X and Y are two points on the string.



What are the directions of motion of X and Y?

[1 mark]

	Direction of motion of X	Direction of motion of Y
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

☐
☐
☐
☐

2 1

Light of wavelength λ is incident normally on a diffraction grating of slit separation 5λ .

The angle between the first-order maximum and the fourth-order maximum on the same side of the central maximum is

[1 mark]

A 11.5° ☐

B 34.5° ☐

C 41.6° ☐

D 53.1° ☐



2 2

A simple pendulum of length 1.2 m oscillates freely with a time period T .

What decrease in length will produce a time period of $\frac{T}{2}$?

[1 mark]

A 5 cm ☐

B 30 cm ☐

C 60 cm ☐

D 90 cm ☐

2 3

The temperature of a metal wire is decreased.

What happens to the resistance of the wire and to the resistivity of the metal?

[1 mark]

	Resistance	Resistivity	
A	decreases	decreases	☐
B	decreases	doesn't change	☐
C	doesn't change	decreases	☐
D	doesn't change	doesn't change	☐

Turn over for the next question



2 4

A progressive wave travels along a stretched string.
The wave has a frequency of 150 Hz and a speed of 30 m s^{-1} .

What is the phase difference between two points on the string that are 5.0 cm apart?

[1 mark]

A zero

☐

B 0.8 rad

☐

C 1.6 rad

☐

D 3.1 rad

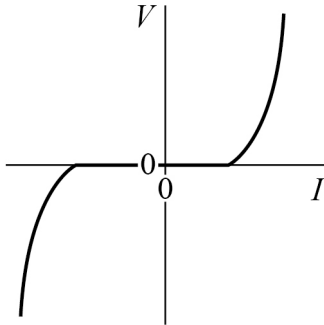
☐

2 5

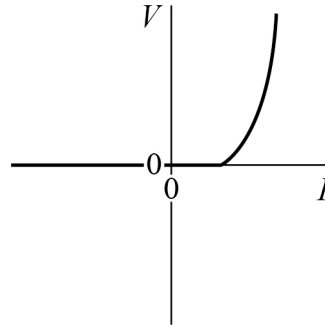
Which is the characteristic of a semiconductor diode?

[1 mark]

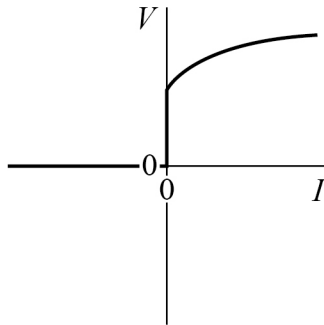
A



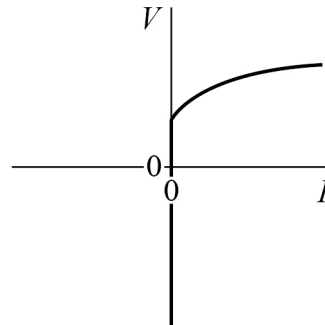
B



C



D



A

☐

B

☐

C

☐

D

☐

[illegible]

[illegible]

[illegible]

01

A mercury atom in a fluorescent tube is excited from its ground state by a collision with an electron.

The mercury atom emits an ultraviolet photon of energy 5.2 eV.

01.1

Calculate the wavelength of the photon.

[2 marks]

wavelength = _____ m

01.2

Explain how this ultraviolet photon causes the production of visible light from the fluorescent tube.

[3 marks]



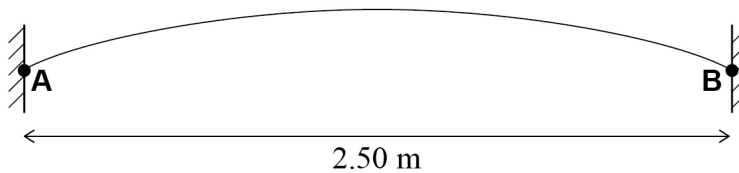
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0 2

A string vibrates between two fixed points **A** and **B**.

Figure 1 shows one position of the string when vibrating at the first harmonic.

Figure 1



The frequency of the first harmonic is 65 Hz.

The tension in the string is 22 N.

The distance **AB** is 2.50 m.

0 2 . 1

Calculate the mass of the string.

[3 marks]

mass = _____ kg

0 2 . 2

The frequency of vibration is increased to form a new stationary wave on the string.

The nodes along the line **AB** are now separated by 0.50 m.

Determine the frequency of the new stationary wave.

[1 mark]

frequency = _____ Hz

4

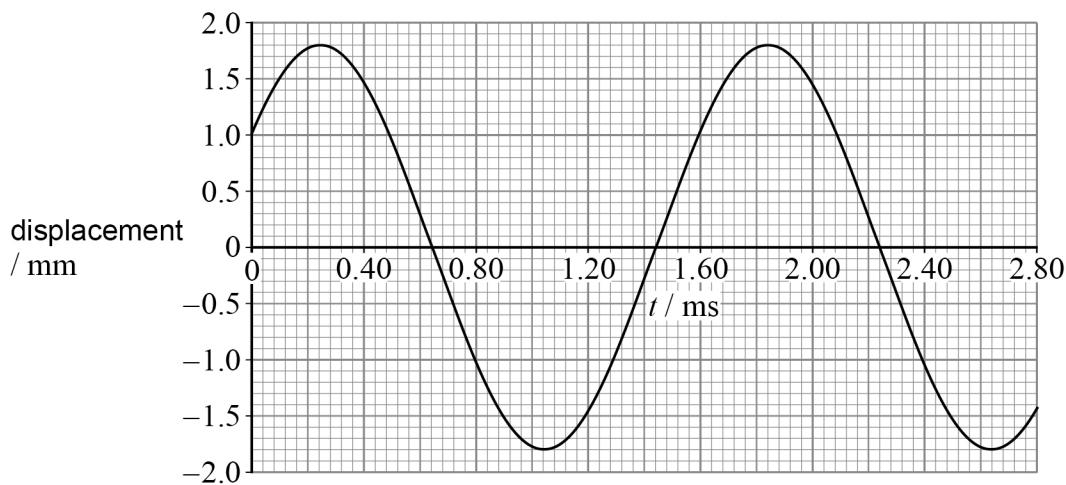


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0 4

Figure 2 shows the variation of displacement of a loudspeaker cone with time t .

Figure 2



0 4 . 1

Determine the total distance travelled by the cone during one complete oscillation.

[2 marks]

distance = _____ mm

0 4 . 2

Determine, in s, the period of oscillation of the cone.

[1 mark]

period = _____ s

0 4 . 3

Deduce the displacement of the cone at $t = 4.00$ ms.

[1 mark]

displacement = _____ mm

0 4 . 4

Calculate the frequency of oscillation of the cone.

[1 mark]

frequency = _____ Hz

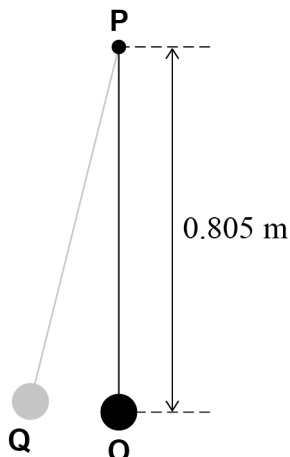


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06

Figure 4 shows a simple pendulum of length 0.805 m suspended from point **P**.

Figure 4



The pendulum is displaced through a small angle from the equilibrium position **O** to **Q**. Moving from **O** to **Q** increases the gravitational potential energy E_p of the pendulum by 7.8 mJ.

The pendulum is released from rest at **Q** at time $t = 0$ and then oscillates freely. The period of the pendulum is 1.80 s.

06.1

Sketch, on **Figure 5**, the variation of kinetic energy E_k of the pendulum with t for one complete oscillation.

Label each axis with a suitable scale.

The time axis on your graph should begin at $t = 0$

[3 marks]

Figure 5

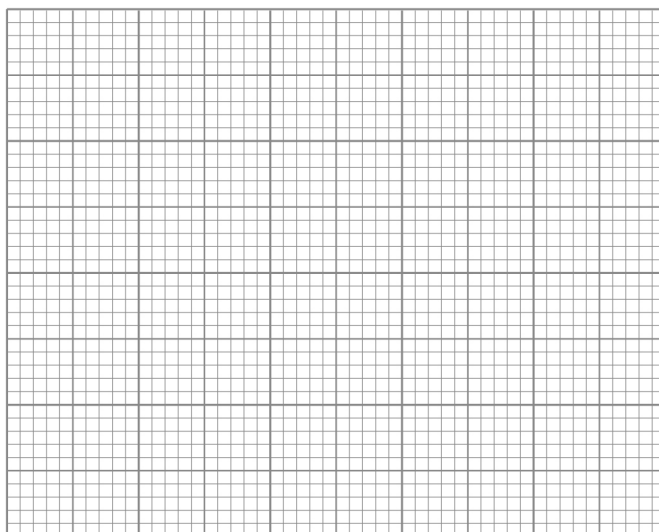
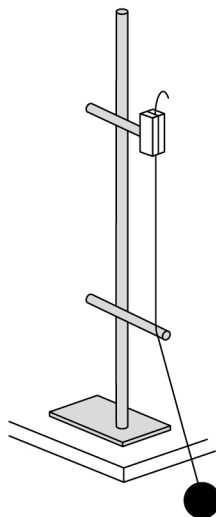
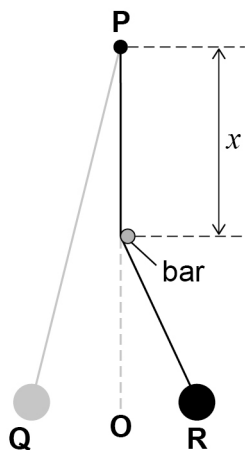


Figure 6 shows a bar placed at a vertical distance x below **P**.

Figure 6

not to scale



The pendulum is again released from rest at **Q**. The string makes contact with the bar. The pendulum reaches its maximum displacement at **R**. No energy transfer occurs between the string and the bar.

When the pendulum is oscillating between **O** and **R**, the pendulum length is $(0.805 - x)$ m. The pendulum takes 1.61 s to move from **Q** to **R** and back to **Q**.

0 6 . 2 Show that the period for a pendulum with length $(0.805 - x)$ m is approximately 1.4 s. **[3 marks]**

0 6 . 3 Calculate x .

[3 marks]

$x =$ _____ m

Turn over ►



0 6 . 4

An energy transfer between E_p and E_k occurs as the pendulum oscillates between **Q** and **O** and as it oscillates between **O** and **R**.

Deduce whether the rate of energy transfer is greater between **Q** and **O** or between **O** and **R**.

[2 marks]

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ANSWER IN THE SPACES PROVIDED**

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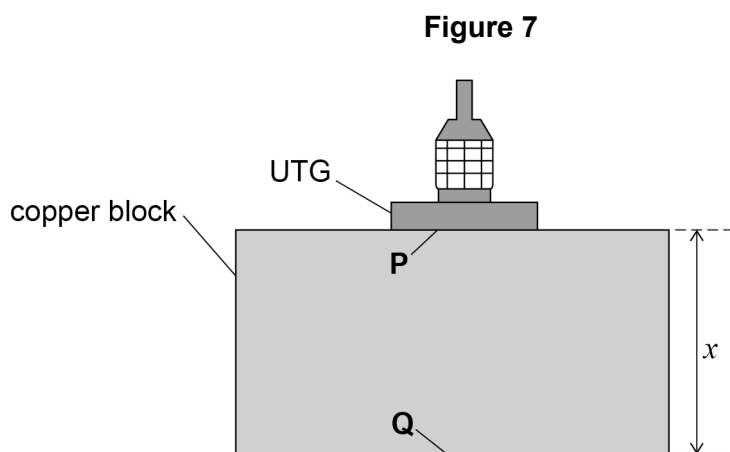
07.1

Describe the nature of a longitudinal wave.

[2 marks]

An ultrasonic thickness gauge (UTG) uses ultrasound waves to determine the thickness of solid materials.

Figure 7 shows the UTG on a copper block of thickness x .



The UTG emits a very short pulse of ultrasound into the block at **P**.

The UTG then detects the waves reflected from **Q**.

Multiple reflections occur at the surfaces of the block as shown in **Figures 8a** and **8b**. The paths of the waves are vertical but are shown at an angle for clarity.

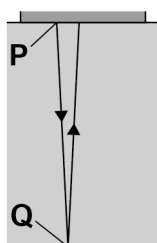
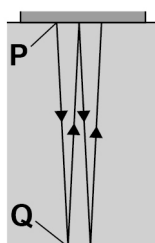
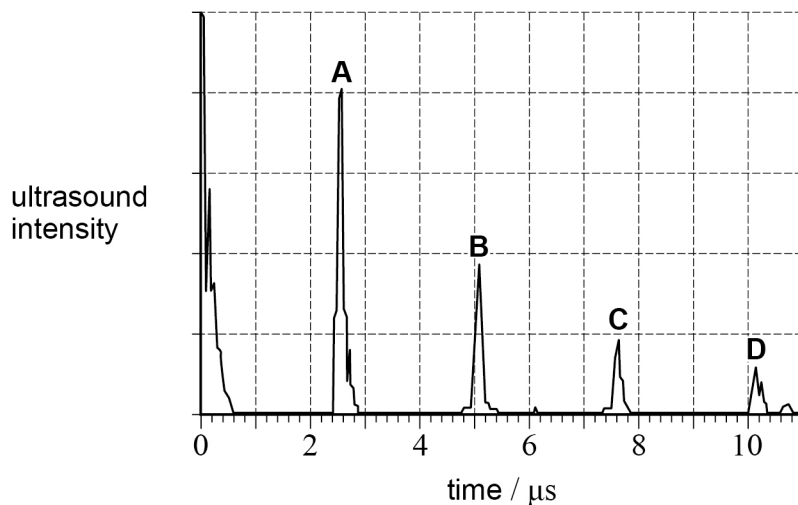
Figure 8a**Figure 8b**

Figure 9 shows the variation of ultrasound intensity with time as detected by the UTG.

Figure 9



Peak **A** is detected when the pulse reflects once at **Q** as shown in **Figure 8a**.

Peak **B** is detected when the pulse reflects twice at **Q** as shown in **Figure 8b**.

Peaks **C** and **D** are produced by further reflections at **Q**.

0 7 . 2

The speed of the ultrasound in the block is 4660 m s^{-1} .

Determine x .

[3 marks]

$x =$ _____ m

Question 7 continues on the next page

Turn over ►



The UTG is used to measure the thickness of a painted copper pipe.

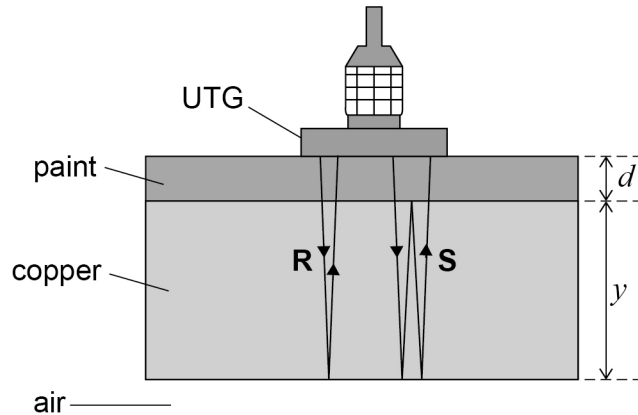
Figure 10 shows the UTG on the outer surface of the pipe.

The figure also shows two paths **R** and **S** taken by the ultrasound.

The paint has a depth d and the pipe wall has a thickness y .

Figure 10

not to scale



The speed of ultrasound in the paint is much less than in copper.

0 7 . 3

An engineer uses the timing for path **R** to determine y .

Explain why the engineer's determination of y is **not** accurate.

[2 marks]

A beam of monochromatic light travelling in air is incident on the surface of a block of glass. The light is incident on the glass at a non-zero angle to the normal.

Which statement is true about the beam after it has been refracted?

[1 mark]

A The direction changes away from the normal.

☐

B The frequency of the light does not change.

☐

C The light travels faster in the glass.

☐

D The wavelength of the light increases in the glass.

☐

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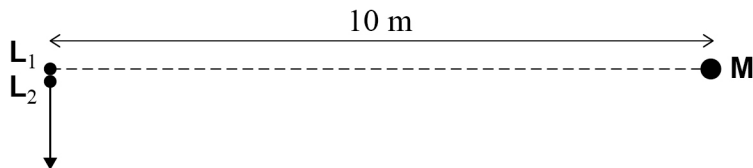
2 **4**

Two loudspeakers L_1 and L_2 emit coherent sound of wavelength 1.0 m.

Initially, L_1 and L_2 are in the same position.

A microphone M at a distance of 10 m from the speakers detects no sound.

L_2 is moved away from L_1 along a line perpendicular to the line L_1M .



What is the smallest distance between L_1 and L_2 that leads to a maximum signal at M ?

[1 mark]

A 0.5 m

B 1.0 m

C 1.6 m

D 3.2 m

[illegible]

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0	1
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A simple pendulum completes 39 oscillations in 1.0 minute.

Calculate the length of the pendulum.

[2 marks]

length = _____ m

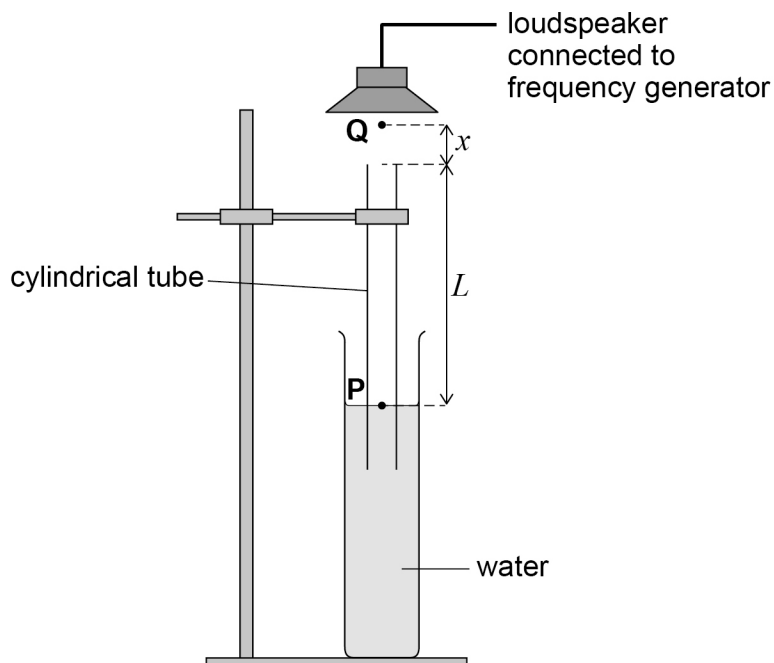
2

07

Figure 7 shows apparatus used to investigate stationary waves in a column of air inside a cylindrical tube.

A loudspeaker, connected to a frequency generator, emits a sound wave with a single frequency into the open end of the tube.

Figure 7



The sound wave from the loudspeaker travels through the air in the tube and reflects at the surface of the water.

07.1

The frequency of the sound is increased from zero. At a certain frequency f , the loudness of the sound in the tube increases significantly.

Explain, with reference to resonance, why the sound is loud at this frequency.

[1 mark]



There is an increase in loudness when a stationary wave forms in the air in the tube.
This stationary wave corresponds to the first harmonic.

Figure 7 shows:

- the position **P** of a node at the surface of the water
- the position **Q** of an antinode.

Q is a constant distance x above the top of the tube.

0 7 . 2 Describe the motion of the air molecules at **P** and at **Q** due to the stationary wave.
[2 marks]

at **P** _____

at **Q** _____

The length L of the column of air is varied.

The frequency f of the first harmonic is recorded for each value of L .

0 7 . 3 When a stationary wave of the first harmonic is formed, the distance **PQ** is equal to $\frac{\lambda}{4}$, where λ is the wavelength of the stationary wave.

Show that f is given by $\frac{1}{f} = \frac{4L}{v} + \frac{4x}{v}$

where v is the speed of sound in air.

[1 mark]

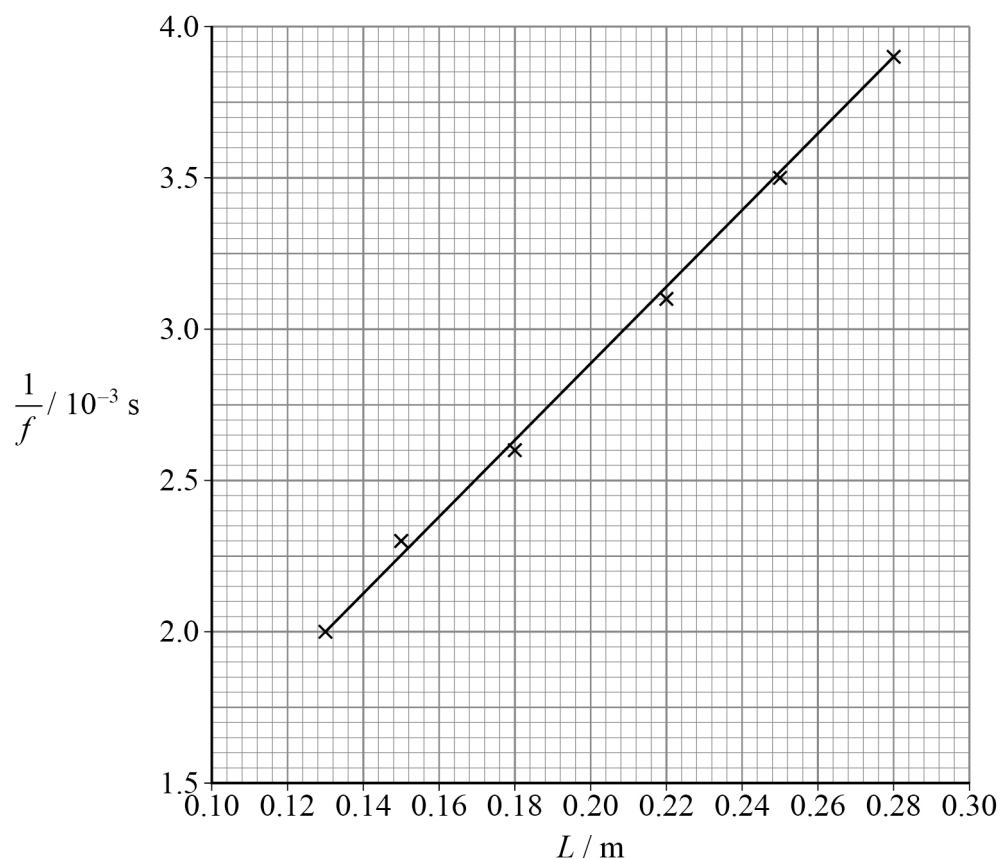
Question 7 continues on the next page

Turn over ►



Figure 8 shows the variation of $\frac{1}{f}$ with L . The best-fit line has been drawn.

Figure 8



0 7 . 4

Determine v using **Figure 8**.

[3 marks]

$v =$ _____ m s^{-1}



0 7 . 5

Explain how you would use **Figure 8** to determine x .
Calculations are not required.

[2 marks]

9

Turn over for the next question

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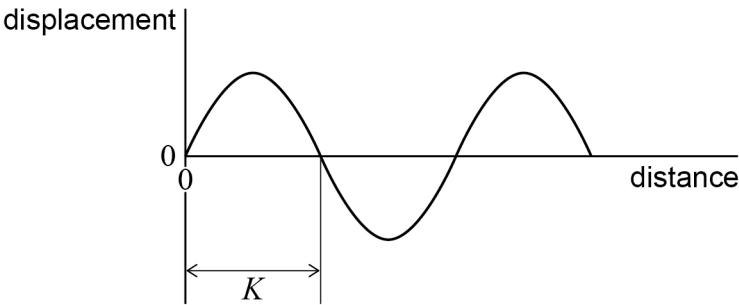


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A transverse wave is created by oscillating a string of beads.

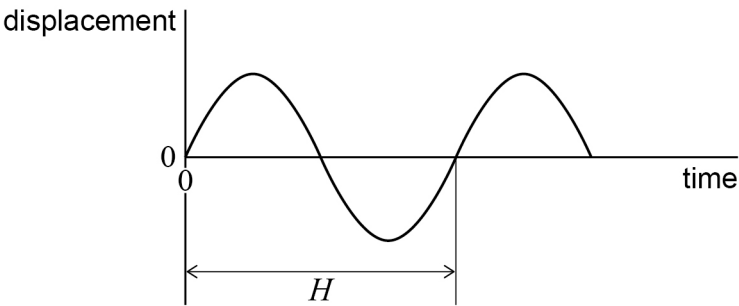
Graph 1 shows the variation of the displacement of each bead with the distance measured from a fixed point along the wave.

Graph 1



Graph 2 shows the variation of the displacement of a particular bead with time.

Graph 2



Which row gives the wavelength and frequency of the wave?

[1 mark]

	Wavelength	Frequency	
A	K	H	☐
B	K	$\frac{1}{H}$	☐
C	$2K$	H	☐
D	$2K$	$\frac{1}{H}$	☐



Don't write

2

0

Sound travels at a speed v in air and at a speed $4.5v$ in water.

A sound wave with a wavelength λ is produced in air. The wave passes from air into water.

Which row shows the wavelength and the nature of the wave in water?

[1 mark]

	Wavelength	Nature	
A	4.5λ	longitudinal	☐
B	λ	longitudinal	☐
C	4.5λ	transverse	☐
D	λ	transverse	☐

The frequency of the first harmonic of a violin string is 440 Hz.

The tension in the string is increased by 8%. The length and mass of the string do not change.

What is the new frequency of the first harmonic of the string?

[1 mark]

A 475 Hz

☐

B 457 Hz

☐

C 452 Hz

☐

D 422 Hz

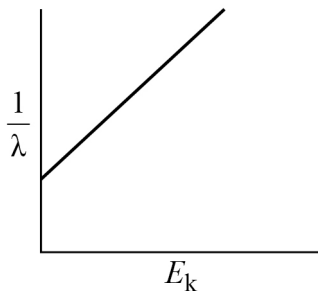
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2 5

Radiation of wavelength λ is incident on a metal surface. Photoelectrons are emitted with maximum kinetic energy E_k .

The graph shows how $\frac{1}{\lambda}$ varies with E_k .



What is the gradient of the graph?

[1 mark]

A h

☐

B hc

☐

C $\frac{1}{h}$

☐

D $\frac{1}{hc}$

☐

[illegible]

[illegible]

Question number	Additional page, if required. Write the question numbers in the left-hand margin.
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Figure 2 shows an experimental arrangement to investigate microwaves.

Figure 2



03

.1

The transmitter produces microwaves of frequency 1.1×10^{10} Hz.

Calculate the wavelength of the microwaves.

[1 mark]

wavelength = _____ m

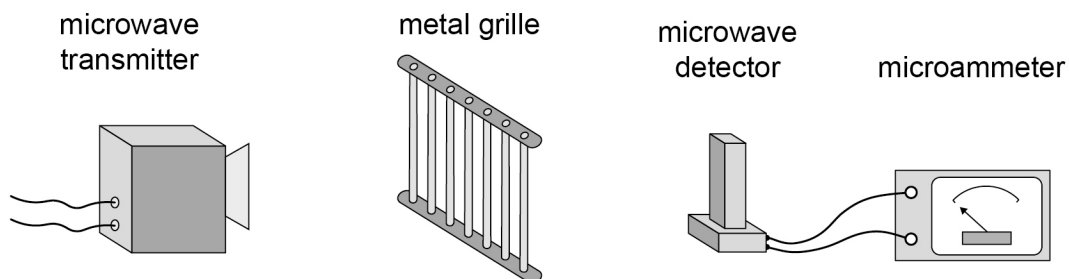


0 3 . 2

A metal grille is placed between the microwave transmitter and the detector as shown in **Figure 3**.

As the grille is placed in position, the microammeter reading decreases to zero.

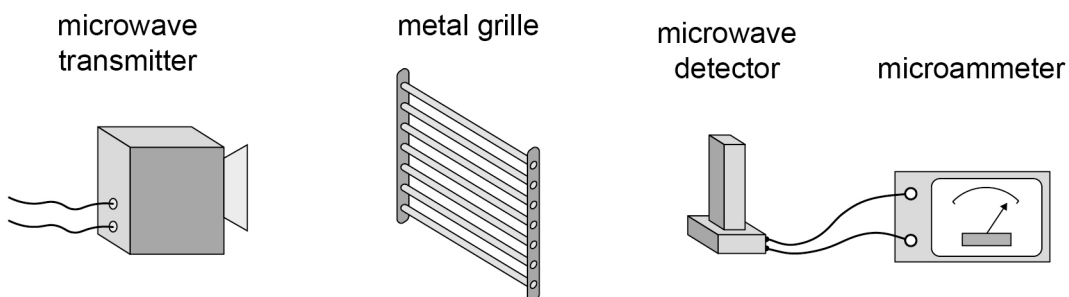
Figure 3



The grille is then rotated in the vertical plane by 90° as shown in **Figure 4**.

As the grille is rotated, the microammeter reading increases to a maximum.

Figure 4



The grille is rotated by a further 90° and the microammeter reading again decreases to zero.

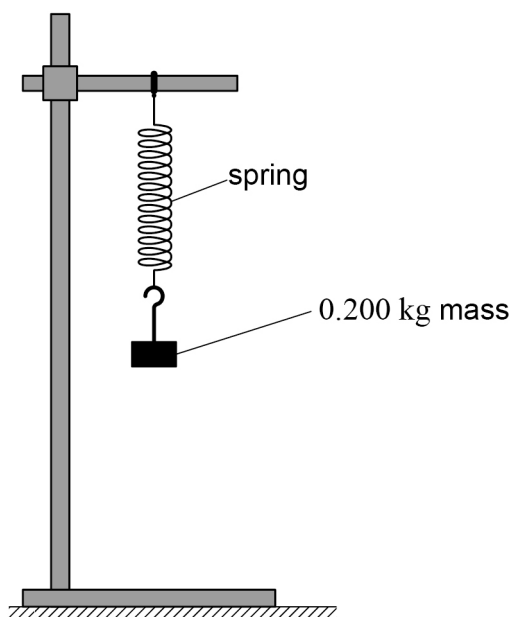
State and explain how these observations provide evidence about the nature of microwaves.

[3 marks]

Do not write

Figure 7 shows a 0.200 kg mass suspended from a spring. The top of the spring is fixed in place.

Figure 7



The mass is displaced vertically downwards and released. The mass–spring system oscillates with a frequency of 1.8 Hz.

Determine the spring constant of the spring.

[2 marks]

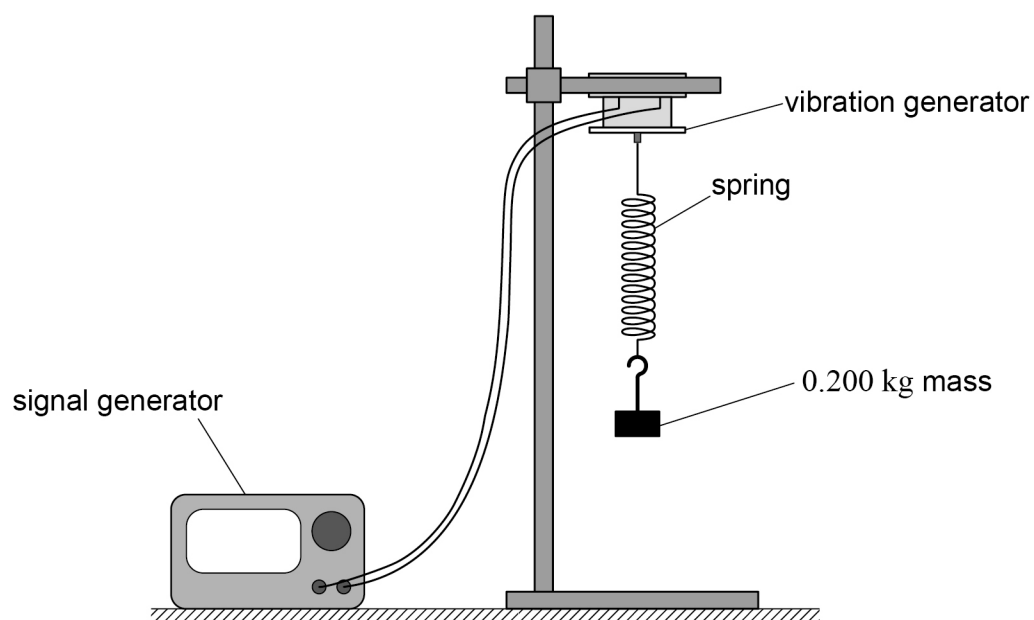
spring constant = _____ N m^{-1}

Question 6 continues on the next page



Figure 8 shows the spring suspended from a vibration generator. The vibration generator is powered by a signal generator.

Figure 8



0 6 . 2

The frequency of the signal generator is increased slowly from zero. The amplitude of oscillation of the mass increases and then decreases. The amplitude has a maximum value at a frequency of 1.8 Hz.

Explain why the amplitude varies in this way.

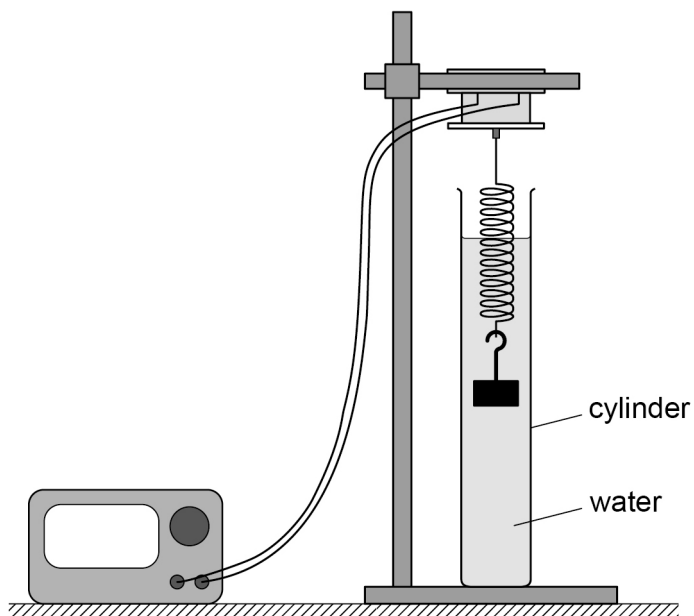
[2 marks]



0 6 . 3

The experiment is repeated with the mass and spring suspended in a cylinder of water as shown in **Figure 9**.

Figure 9



The frequency is again increased slowly from zero.

Compare the behaviour of the oscillating systems in **Figure 8** and **Figure 9**.

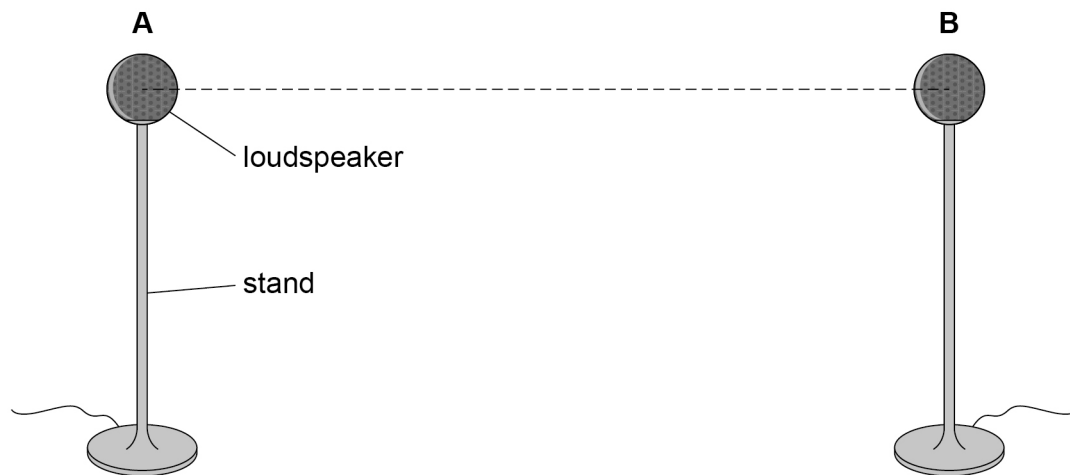
[2 marks]

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Figure 10 shows two small loudspeakers **A** and **B** on stands. **A** and **B** act as point sources of sound waves.

Coherent waves of the same amplitude are emitted from **A** and **B**.

Figure 10



State what is meant by coherent waves.

[2 marks]



A sound meter gives a maximum reading when placed at the centre of the line **AB**. When the meter is moved 5.0 cm towards **B** along the line **AB**, the meter reading decreases to a minimum.

0 8 . 2

Explain why the meter gives a minimum reading at this new position.

[3 marks]

As the meter is moved further towards **B** another maximum is detected, followed by a regular pattern of minima and maxima.

0 8 . 3

State the distance between two adjacent minima.
Go on to explain your answer.

[2 marks]

distance = _____ m

0 8 . 4

As the meter gets closer to **B**, the amplitude of the minima increases.

Suggest why.

[2 marks]

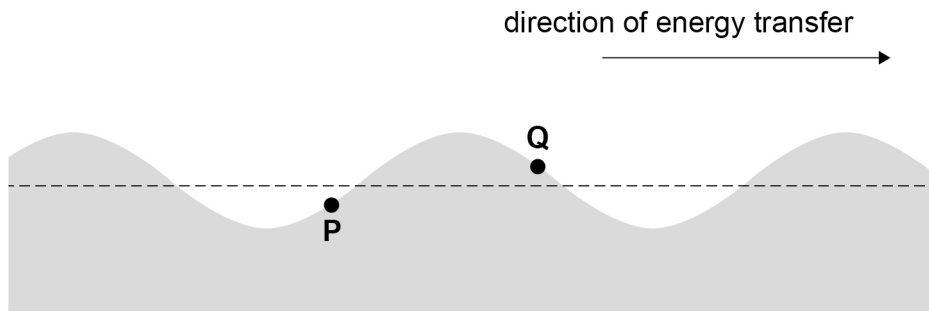


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1

2

The diagram shows a progressive transverse wave at one instant.
The direction of energy transfer is to the right.



What are the directions of movement of **P** and **Q** at this instant?

[1 mark]

	Direction of movement of P	Direction of movement of Q	
A	down	down	☐
B	down	up	☐
C	up	down	☐
D	up	up	☐



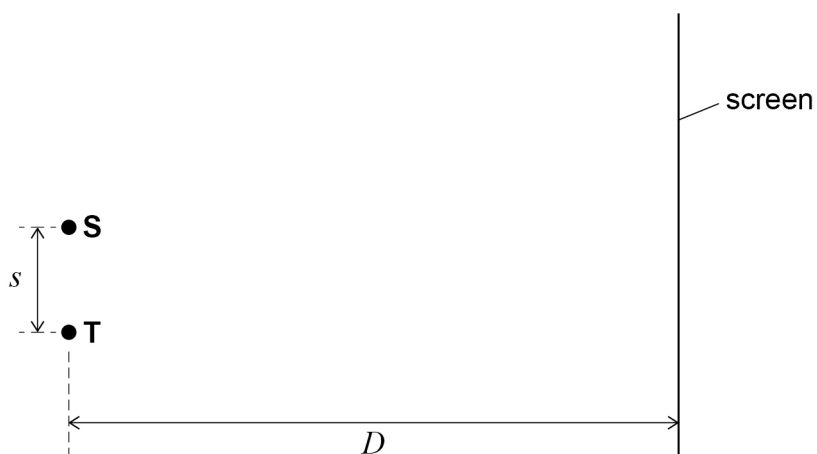
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Two coherent sources **S** and **T** emit light of wavelength λ .

S and **T** are separated by a distance s .

A screen lies at a distance D from a line connecting **S** and **T**.

In this arrangement $s \ll D$.



The distance between the 5th and 7th bright fringes is:

[1 mark]

A $\frac{\lambda D}{2s}$ ☐

B $\frac{2\lambda D}{s}$ ☐

C $\frac{3s\lambda}{D}$ ☐

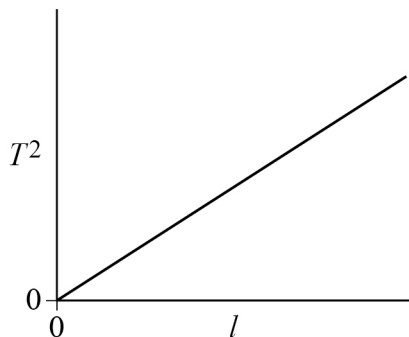
D $\frac{3\lambda D}{s}$ ☐

Turn over for the next question



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A simple pendulum is used to find a value for the acceleration g due to gravity. The period of oscillation T is measured for different lengths l of the pendulum. A graph of T^2 against l is a straight line with gradient S . The graph goes through the origin.



What is g ?

[1 mark]

A $\frac{2\pi}{S}$ ☐

B $2\pi S$ ☐

C $\frac{4\pi^2}{S}$ ☐

D $\frac{S}{4\pi^2}$ ☐

Monochromatic light has a wavelength λ .
Each photon of the light has an energy E .

Which row shows what happens to λ and E as the light enters a glass block from air?

[1 mark]

	λ	E	
A	decreases	remains the same	☐
B	decreases	increases	☐
C	increases	remains the same	☐
D	increases	decreases	☐



Do not write

A stretched wire vibrates between two fixed points.

The frequency of the first harmonic of the vibrating wire is 400 Hz.

The tension in the wire is T .

The tension in the wire is now increased.

Changes to the length and the mass per unit length of the wire are negligible.

The frequency of the first harmonic is now 600 Hz.

What is the new tension in the wire?

[1 mark]

A $\sqrt{\frac{3T}{2}}$ ☐

B $\sqrt{\frac{3}{2}} T$ ☐

C $\frac{3T}{2}$ ☐

D $\frac{9T}{4}$ ☐

2**2**

The phase difference between two points on a wave is 0.56 rad.

What is the phase difference in degrees?

[1 mark]

A 32°

☐

B 64°

☐

C 101°

☐

D 202°

☐

Turn over ►



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2**3**Do not write
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box

The energy storage capacity of a battery is rated in ampere hour (A h).
A battery with a storage capacity of 1 A h can provide 1 A for one hour at its working pd.

One battery has a capacity of 2700 mA h at a working pd of 4.2 V.

What is the energy that can be transferred by this battery?

[1 mark]

A 11 kJ

☐

B 41 kJ

☐

C 680 kJ

☐

D 41 MJ

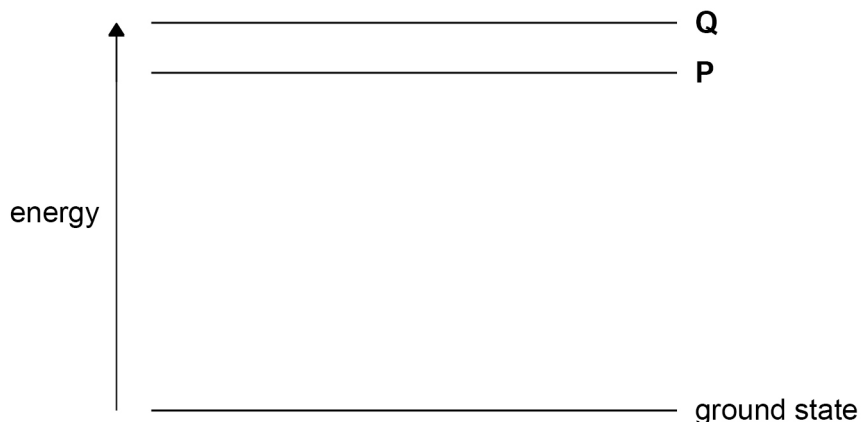
☐

2 **5** The diagram shows the ground state of an atom and two excited states **P** and **Q**.

outside the
box

The transition of an electron from **P** to the ground state produces a photon of wavelength 502 nm.

The transition of an electron from **Q** to the ground state produces a photon of wavelength 448 nm.



The energy difference between **P** and **Q** is:

[1 mark]

A $4.8 \times 10^{-20} \text{ J}$

B $4.0 \times 10^{-19} \text{ J}$

C $4.4 \times 10^{-19} \text{ J}$

D $3.7 \times 10^{-18} \text{ J}$

[illegible]

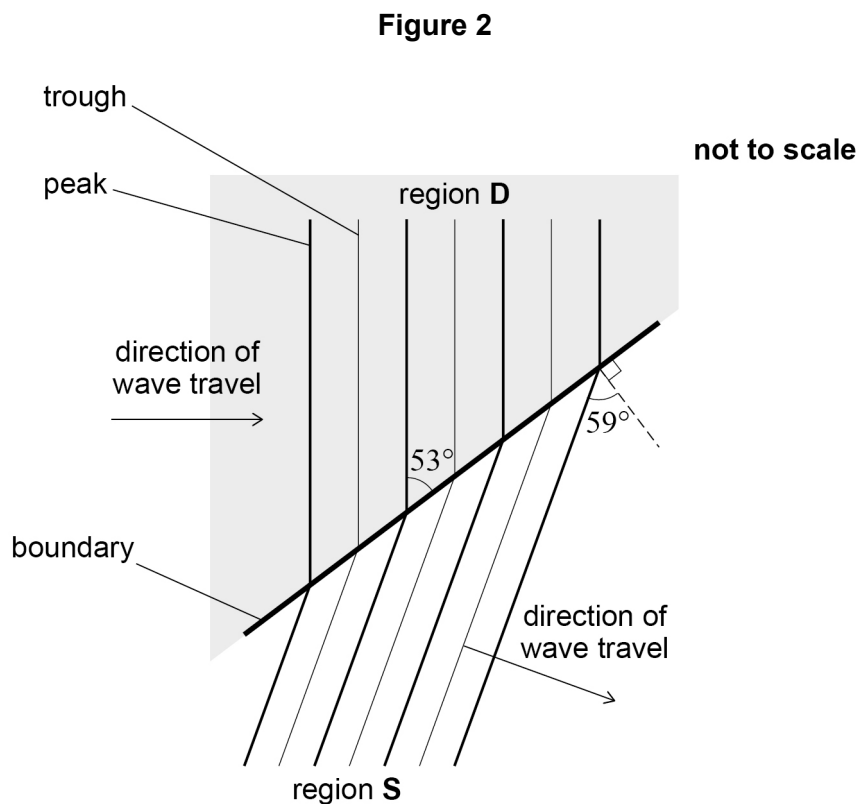


[illegible]

03

A wave on the surface of water is used to model the behaviour of light waves.

Figure 2 shows the wave travelling on the surface of water, as viewed from above. The boundary between a deep region **D** and a shallow region **S** is shown.



The wave travels more slowly in shallow water than in deep water.

03.1

Explain why the wavelength of the wave is greater in region **D** than in region **S**.

[1 mark]



0 3 . 2 The speed of the wave in region **D** is v_D . The speed of the wave in region **S** is v_S .

The ratio of the speeds is given by:

$$\frac{v_D}{v_S} = \frac{\sin \theta_S}{\sin \theta_D}$$

where θ_D is the wave's angle of incidence in region **D**

θ_S is the wave's angle of refraction in region **S**.

Calculate $\frac{v_D}{v_S}$.

[1 mark]

$$\frac{v_D}{v_S} = \underline{\hspace{10em}}$$

2

Turn over for the next question



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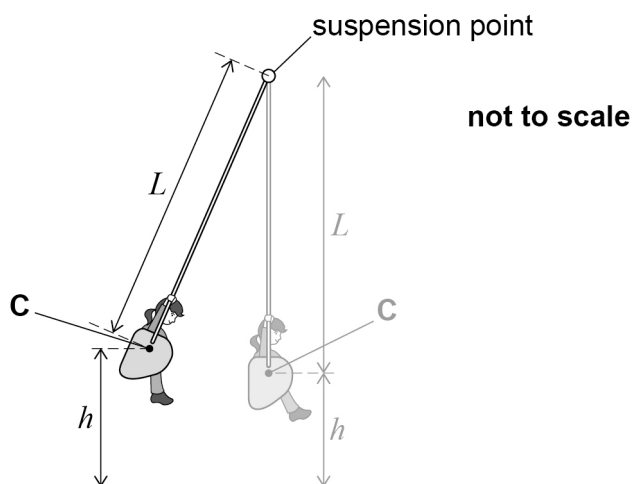
This question is about a child moving on a swing.

C is the centre of mass of the child and seat.

h is the height of **C** above the ground.

Figure 3 shows **C** at two different values of h .

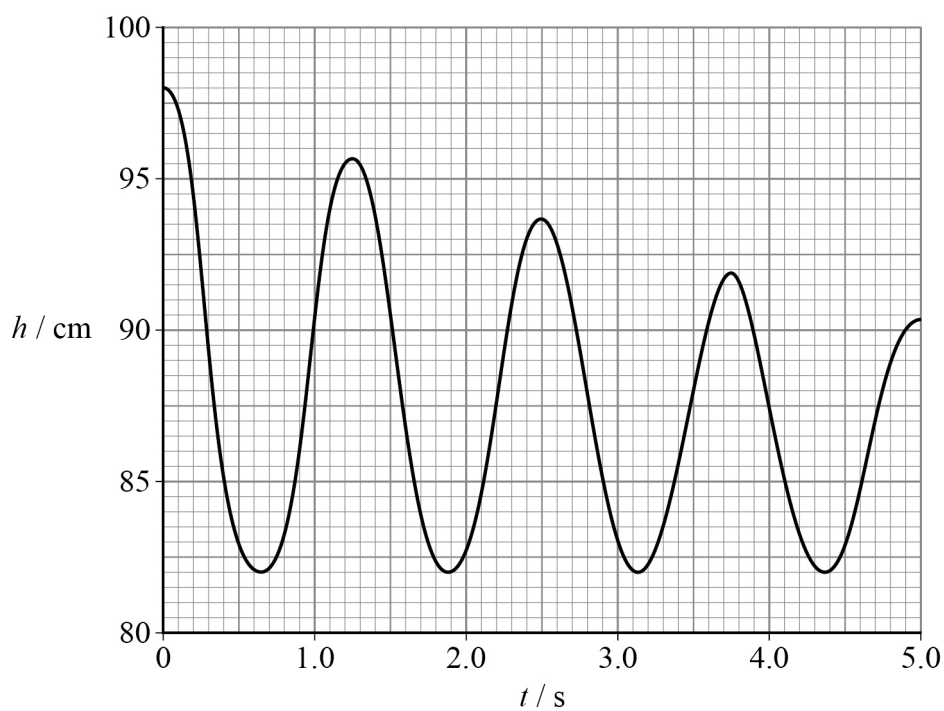
Figure 3



At time $t = 0$ the child starts swinging.

Figure 4 shows the variation of h with t .

Figure 4



0 4 . 1 Determine the time period of the oscillation.

Go on to explain how you used **Figure 4** to determine your answer.

[2 marks]

time period = _____ s

0 4 . 2 L is the distance from **C** to the suspension point of the swing.

Assume that the time period of the swing is the same as the time period for a simple pendulum of length L .

Estimate L .

[2 marks]

L = _____ m

Question 4 continues on the next page

Turn over ►



0 4 . 3

Suggest **one** reason why the swing does not behave in exactly the same way as a simple pendulum.

[1 mark]

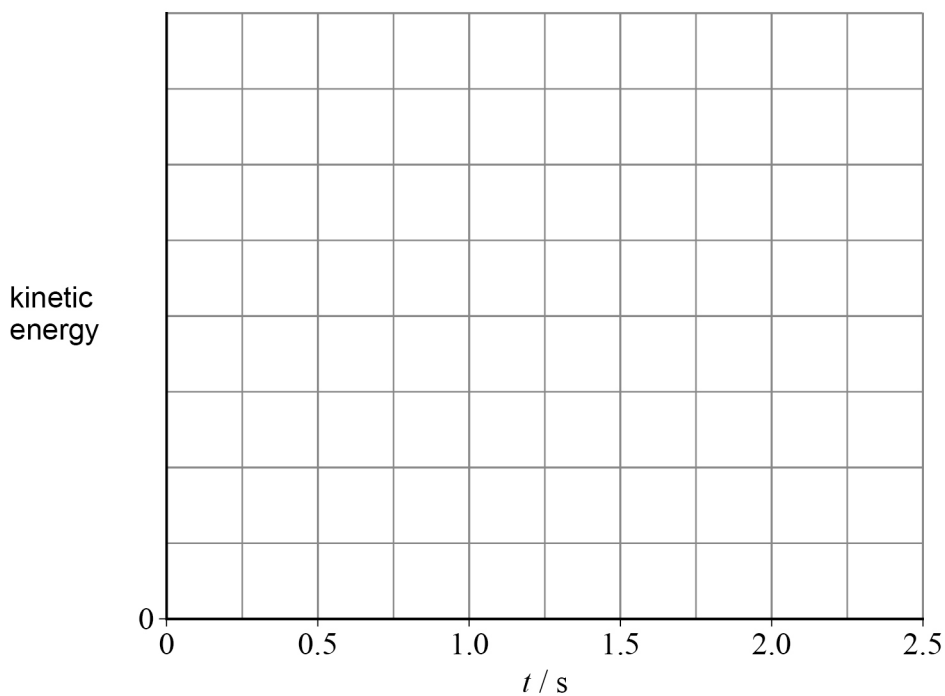
0 4 . 4

Sketch, on **Figure 5**, the variation of the child's kinetic energy with t for the first 2.5 s of the motion.

Do not add values to the kinetic energy axis.

[3 marks]

Figure 5



Do not write

0 5 . 1

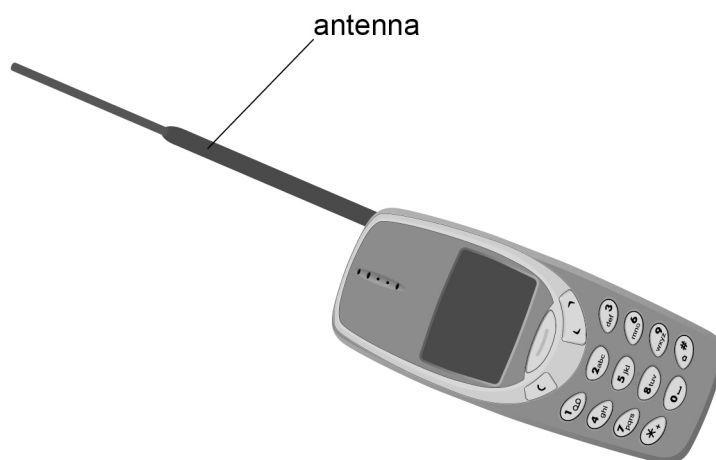
Explain why longitudinal waves cannot be polarised but transverse waves can be polarised.

[2 marks]

0 5 . 2

Figure 6 shows an early form of mobile phone. The antenna on this mobile phone was a long thin metal rod that received microwave signals.

Figure 6



Explain why the microwaves transmitted to the mobile phone must **not** be polarised in a single plane.

[2 marks]



Do not write

Describe **two** differences between the properties of a stationary wave and the properties of a progressive wave.

[2 marks]

1 _____

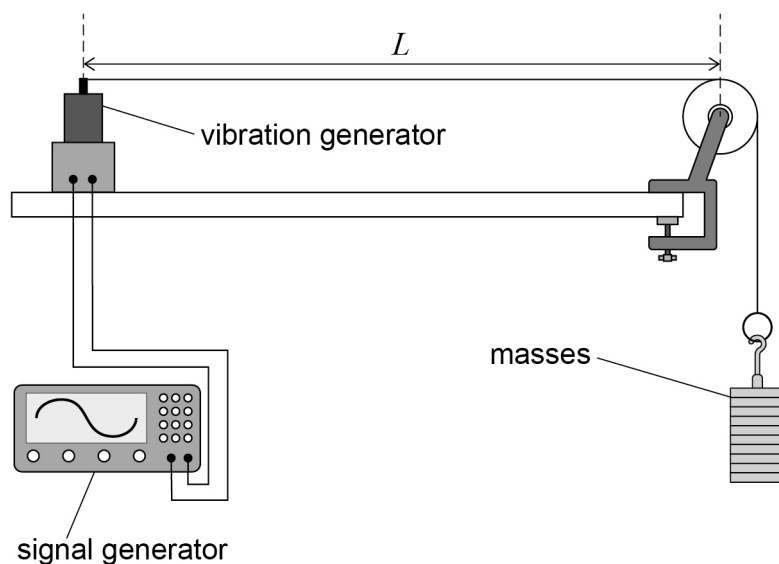
2 _____

Question 8 continues on the next page



Figure 11 shows the apparatus that a student uses to investigate stationary waves on a string.

Figure 11



The string is attached to a vibration generator at one end and hangs over a pulley at the other end. A selection of different masses is available to change the tension in the string.

The vibrating length L of the string cannot be changed.

The frequency of the vibration generator is increased from zero until the first-harmonic stationary wave is observed on the string.

0 8 . 2

Describe a suitable method to determine the mass per unit length μ of the string using this apparatus.

In your answer, include:

- the measurements to be taken, including details of the measuring instruments used
- techniques to reduce the uncertainty in the experiment
- how μ is determined using a graphical method.

[6 marks]



When the student repeats the experiment, his measurements of the first-harmonic frequency vary at each tension. This is partly due to a random error in identifying the first harmonic.

[1 mark]

Don't write

An oscillating system consists of an object of mass M suspended from a spring of stiffness k .

The object is displaced vertically and released. The system oscillates freely with a frequency f .

An object of mass $2M$ is suspended from a spring of stiffness $3k$.

What is the frequency of oscillation of this system?

[1 mark]

A $1.5f$

☐

B $1.2f$

☐

C $0.82f$

☐

D $0.67f$

☐

Do not write

A transverse wave travels along a string.

The transverse wave has a frequency of 2.0 Hz, a wavelength of 4.0 cm and an amplitude of 3.0 cm.

What is the total distance travelled by a point on the string in one second?

[1 mark]

A 6.0 cm

☐

B 8.0 cm

☐

C 12 cm

☐

D 24 cm

☐

Two aerials $\mathbf{A}_1$ and $\mathbf{A}_2$ receive radio waves from the same distant transmitter $\mathbf{T}$.

The distance between $\mathbf{A}_1$ and $\mathbf{T}$ is d_1 .

The distance between $\mathbf{A}_2$ and $\mathbf{T}$ is d_2 .

The waves have a frequency of 104 MHz.

The phase difference between the waves received by $\mathbf{A}_1$ and $\mathbf{A}_2$ is 1.4 rad.

What is the minimum value for $d_1 - d_2$?

[1 mark]

A 0.32 m

☐

B 0.64 m

☐

C 1.3 m

☐

D 13 m

☐

Don't write

09

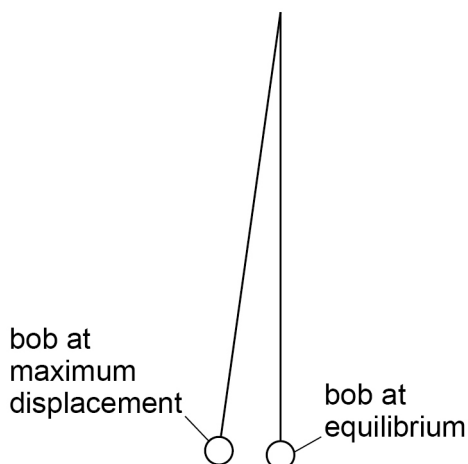
Figure 10 shows a simple pendulum used by a student to determine the acceleration g due to gravity.

The pendulum is displaced and allowed to swing freely. The time taken for one oscillation is T .

The student intends to use a stopwatch to measure $10T$, the time for 10 oscillations, for several different lengths of pendulum.

The positions of the pendulum bob at equilibrium and at maximum displacement are shown.

Figure 10



09.1

Explain why the amplitude of the oscillation decreases with time.

[1 mark]



0 9 . 2

The student uses a fiduciary marker to help him measure $10T$.

Annotate **Figure 10** with an **X** to show a suitable position for the fiduciary marker.

Go on to explain your choice of position.

[2 marks]

0 9 . 3

Explain the advantage of measuring $10T$ rather than directly measuring T .

[2 marks]

0 9 . 4

The student makes a mistake. He measures the time taken for 9 oscillations for each length, but records these times as $10T$.

Deduce the value of g obtained by the student.

[2 marks]

value of $g =$ _____ m s^{-2}

END OF SECTION A



Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

1 1 . 1

Explain how antinodes are formed in a stationary wave.

[3 marks]

Question 11 continues on the next page

Turn over ►

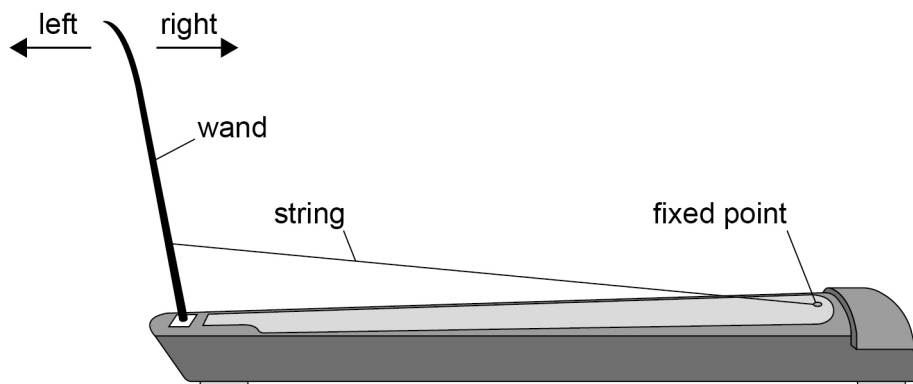


Figure 13 shows a single-stringed musical instrument.

The string is fixed at one end and the other end of the string is attached to a long wooden rod called the wand.

The wand can be moved slightly to the left or right and the string is plucked (pulled quickly and then released) to make a sound.

Figure 13



When the wand is in its equilibrium position, and the middle of the string is plucked, the string vibrates at its first harmonic.

The frequency of the first harmonic is 131 Hz.

The length of the string is 0.84 m.

The mass per unit length of the string is 0.015 g cm^{-1} .

1	1
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.

2

 Calculate the tension in the string when the wand is in the equilibrium position.

[2 marks]

tension = _____ N

1 1 . 3

When the string is plucked at a certain position, it creates an antinode at that position on the string.

When the string is plucked at a distance x from the wand, the second harmonic is heard.

Calculate x .

[1 mark]

$x =$ _____ m

1 1 . 4

The wand is pulled to the left and held. This produces a very small increase in the length of the string.

The middle of the string is plucked and the first harmonic is produced. The first harmonic now has a significantly higher frequency than 131 Hz.

Explain why a very small increase in the length of the string leads to a large increase in the frequency of the sound.

[2 marks]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

1	7
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A mass–spring system oscillates with frequency f .

The mass is increased by 21%.

What is the frequency of the new oscillation?

[1 mark]

A $0.83f$ ☐

B $0.79f$ ☐

C $0.91f$ ☐

D $1.1f$ ☐



Do not write

A spring of length 4.0 m is used to demonstrate a longitudinal wave.

One end of the spring is moved back and forth. One oscillation is completed in a time of 0.40 s.

A compression produced by this movement takes 2.0 s to travel the complete length of the spring.

What is the wavelength of the wave?

[1 mark]

A 0.40 m

☐

B 0.80 m

☐

C 1.6 m

☐

D 2.0 m

☐

Two loudspeakers are used to demonstrate sound wave interference.

So that regions of maximum and minimum loudness are observed, the sounds from both speakers **must**:

[1 mark]

A be directed towards each other.

☐

B have the same amplitude as each other.

☐

C be coherent with each other.

☐

D be directed parallel to each other.

☐

Turn over ►



Do not write